WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST

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[TODO: Abstract to be written after the application.]

1. Introduction. [TODO: Introduction (not drafted yet).]

2. Data.

2.1. *Conflict and unrest events.* Event data come from the Armed Conflict Location & Event Data Project (ACLED; Raleigh et al., 2010). ACLED records the date, location, actors and type of individual events of political violence and demonstrations, compiled from news media, local partners and other sources. We use events from 1 January 2018 to 31 December 2024 and group ACLED event types into two outcomes: *violent conflict* (battles, explosions/remote violence and violence against civilians) and *social unrest* (riots and protests). Events are aggregated to a regular $0.5^\circ \times 0.5^\circ$ longitude–latitude grid, the resolution of PRIO-GRID (Tollefsen, Strand and Buhaug, 2012), and to calendar weeks. For cell s and week t, let $N_{st}^{(j)}$ be the number of events of type j, and let

$$Y_{st} = \mathbf{1}\Big\{\textstyle\sum_{j\in\mathcal{J}} N_{st}^{(j)} > 0\Big\},$$

where $\mathcal{J}$ is the set of event types of the outcome being modelled. The two outcomes are analysed separately.

The main analysis covers Africa, the Middle East and adjacent regions, the box from 20°W to 65°E and from 36°S to 45°N, with 13,719 cells in 93 countries. We call it the AME region. Besides Africa and the Middle East, the box includes Mediterranean Europe, the Caucasus, parts of Central Asia and Afghanistan. ACLED coverage of these areas has a different history. Among the countries in the box, Italy, Spain, Portugal, France, Malta and Andorra are covered by ACLED only from 2020 (as are several small islands of the Indian and Atlantic oceans), so they contribute three of the five training years, 2020–2022, which include the period of COVID-19 restrictions.

Event weeks are rare. In 2023–2024, over the covered cell-weeks of the AME region with complete history, 2.4% of cell-weeks contain at least one conflict event and 2.7% at least one unrest event. Over all covered cell-weeks worldwide with complete climate records the shares are 1.6% and 2.7%.

Figure 1 shows where and when events are recorded in the AME region. Over 2018–2024, 2.3% of covered cell-weeks in this region contain at least one conflict event and 2.3% at least one unrest event. Worldwide, the shares are 1.3% and 2.1%. Events are concentrated in space:

- a third of the cells in the region (33.6% for conflict, 35.7% for unrest) record any event at all;

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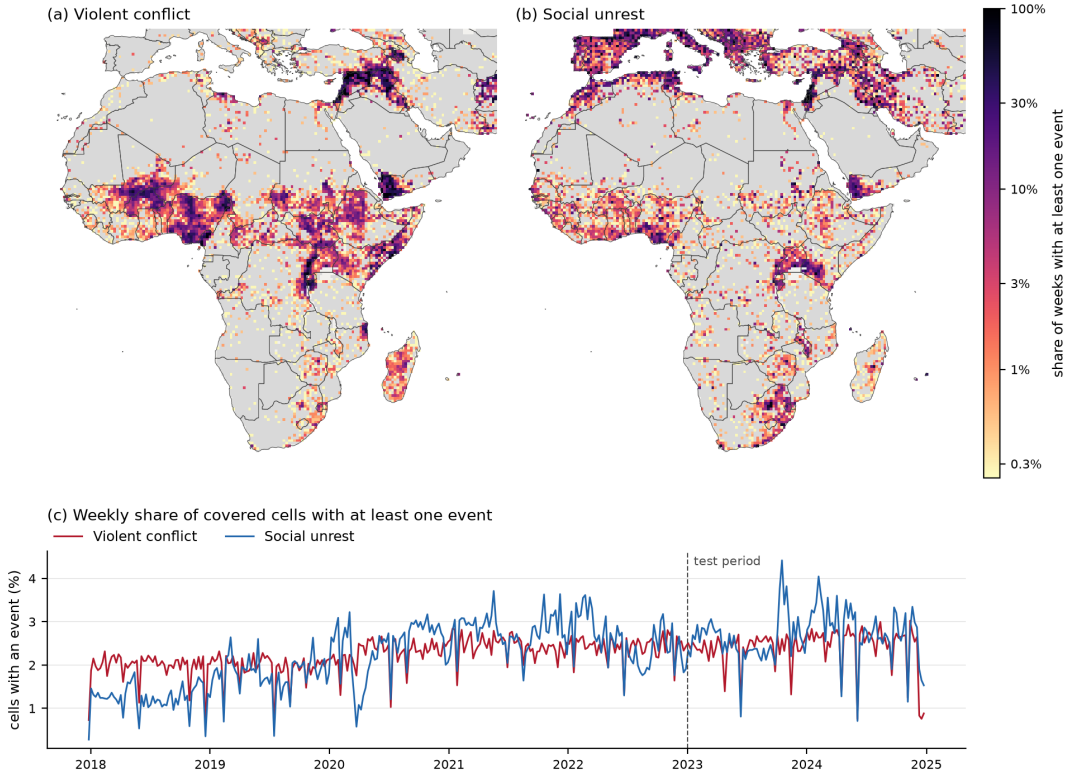


FIG 1. *ACLED events on the* 0.5° *grid in the AME region (Africa, the Middle East and adjacent regions), 2018–2024, over covered cell-weeks. (a, b) Share of weeks with at least one event in each cell, for violent conflict and social unrest (logarithmic colour scale); grey cells have no event week. (c) Weekly share of covered cells in the region with at least one event. The dashed line marks the start of the test period.*

- the 1% of cells with the most event weeks account for 30% of all conflict event weeks and 27% of all unrest event weeks;
- fewer than 1% of cells record events in more than half of all weeks. These cells, for example in Syria, Yemen, northeastern Nigeria and eastern Democratic Republic of the Congo, are one reason why recent history is such a strong predictor (Section 2.2).

The two outcomes have different geographies. Conflict forms large contiguous regions: the central Sahel and Lake Chad basin, the Sudans and Ethiopia, Somalia, eastern Congo, Yemen and Syria–Iraq. Unrest is densest in Mediterranean Europe, North Africa, Turkey, Iran, South Africa and the Great Lakes region, where population and media coverage are high. The weekly share of cells with an event rises over the period. For conflict it goes from about 2% in 2018 to about 2.5% in 2024, and for unrest from about 1.3% to between 2% and 4%. The test period is therefore not exchangeable with the early training years. The series also has isolated weeks with far fewer events than the neighbouring weeks. We checked whether they arise in our construction of the table. No week is split between the partitions of the source table, every cell-week appears once, and the number of cells is constant in the weeks around each of them. The dips are therefore present in the weekly counts of the source partitions, and we keep them as recorded. [TODO: Sensitivity of the main results to excluding these weeks.]

The grid consists of all 0.5° cells that intersect a country polygon of Natural Earth (1:10m), excluding cells without a terrestrial biome (rock and ice, and large lakes). Each cell is assigned to the country that contains its centroid. A coastal cell whose centroid falls in the sea

is assigned to the nearest country. Every cell-week appears exactly once, and national indicators are attached through the assigned country, with the exceptions described below. The result is 59,339 cells observed for up to 366 weeks.

The table built by the original pipeline differs in one important respect. About 8.5% of its 34.7 million distinct cell-weeks (before the biome exclusion) occur more than once, with multiplicities such as 2^k, 3^k and $2 \cdot 3^k$, the signature of many-to-many joins between the grid and polygons of countries, IPC regions and national indicators. These copies make up 50.5% of its 70.0 million rows. The copies of a cell-week share its events but often disagree on the covariates. Among duplicated cell-weeks, the copies differ in the EPI and the liberal democracy index in 62% of cases, in GDP in 49%, in the IPC phases in about 25% and in the climate variables in 7–11%. We do not know why the climate values of a cell differ between copies; the original pipeline does not record which polygon each copy comes from. We keep the whole first row of each cell-week, so that all covariates of a row come from the same copy. The climate values of duplicated cell-weeks are therefore those of an arbitrary copy, and this part of the error is retained rather than repaired. [TODO: Sensitivity of the main results to using the other copy's climate values, or climate re-extracted from CHIRTS and CHIRPS.] IPC phase populations are the exception: they are averaged over the copies. Five national indicators (EPI, liberal democracy index, greenhouse gas emissions, material and ecological footprint) are re-attached for each country and year from the cells of that country that were not duplicated. When a country-year has no value in these cells, which happens when the indicator is missing for that country, the code falls back to the most common value of the kept first copies of that country-year. Since every country has non-duplicated cells, such values come from copies joined to a neighbouring country's polygon. The fallback therefore can give a country its neighbour's value and turns a missing value into an observed one, so the missingness indicator is not set. In the whole repaired table, 14.5% of rows take at least one national indicator from this fallback. Access to electricity is not re-attached at all: every cell-week keeps the value of its first copy, which in cells that had copies can come from a neighbouring country. It is therefore not constant within a country-year, and its decomposition into between- and within-country parts (Section 3.7) mixes differences across borders with change over time. GDP per capita is a national quantity: World Bank GDP of the country divided by the sum of the GHSL population of its cells, and it is constant within each country-year. Section 5.1 describes the consequences of the duplication for the original analysis.

Two features of ACLED matter for modelling. First, geographic coverage expanded over time, and several countries entered the dataset after 2018. Cell-weeks in a country before its coverage start are *unobserved*, not event free, so they are removed rather than coded as zeros. Coverage start dates are taken from ACLED's country coverage table. Territories not listed there take the date of the country or region that contains them. Before the biome exclusion, coverage removes 7,078,822 cell-weeks in 79 countries and territories; 3,998,319 of them are in Antarctica, which the biome exclusion would remove in any case. The event history is treated as incomplete in the first four covered weeks. After both exclusions the table has 19,364,639 cell-weeks. Second, the probability that an event is recorded depends on media and partner density, so the outcome combines occurrence and reporting (Weidmann, 2016). The two are not separately identified without auxiliary information. The latent fields of Section 3 absorb spatially smooth reporting differences together with unmeasured drivers of risk, and we interpret them accordingly. Reporting intensity also changes over time and by country, for example when ACLED adds sources or partners. The year effects are global, and at test time they are held at their 2022 value, so they cannot absorb country-specific changes in reporting. Such changes may explain part of the rise in recorded unrest above, and they can inflate both the estimated state dependence and the persistence of the space-time fields.

ACLED also codes the spatial and temporal precision of each event. An event whose exact location is unknown is placed at the centroid of the town or administrative unit named in the source, and an event whose exact date is unknown at a nominal date within the reported period. The data available to us are weekly counts by cell, and they carry no precision codes, so we cannot filter or weight events by precision. Imprecisely located events accumulate in the cells that contain administrative centroids and capitals. This can inflate both the persistence of events in those cells and the cell-level latent effects estimated there, and it is one more reason to read those effects as recorded rather than actual risk.

2.2. *Recent event history.* For each cell and week we compute the number of events of each type in the four preceding weeks, $L_{st}^{(j)} = \sum_{l=1}^{4} N_{s,t-l}^{(j)}$, using only past weeks of the same cell. In the models of Section 3 the history enters through $h_{st} = \log\bigl(1 + \sum_{j \in \mathcal{J}} L_{st}^{(j)}\bigr)$ for the outcome being modelled, and through the corresponding quantity for the other outcome as an additional covariate. Cell-weeks whose four-week history is incomplete, at the start of the series and of each country's coverage, are dropped from the parametric latent models. The boosting models, and the boosting offset of the hybrid models, keep these rows and receive the incomplete history as missing.

2.3. *Environmental, socioeconomic and national covariates.* Covariates are listed in Table 1. They vary at three resolutions, which is central to the identification arguments of Section 3.7:

1. *cell–week*: daily minimum and maximum temperature, heat index and wet-bulb globe temperature (CHIRTS), precipitation (CHIRPS), $PM_{2.5}$ (CAMS) and a drought index (Global Drought Observatory);
2. *cell*: population (GHSL), river length and lake area (Natural Earth), malaria risk (Malaria Atlas Project), terrestrial biome (RESOLVE Ecoregions) and, in the boosting models only, an indicator of $|\mathrm{latitude}| < 30^{\circ}$;
3. *country–year*: GDP per capita, access to electricity, the Environmental Performance Index (EPI), the V-Dem liberal democracy index, and per-capita greenhouse gas emissions, material footprint and ecological footprint.

Food insecurity, the population in each IPC phase, is available for subnational regions and periods and is apportioned to cells by population share.

The climate variables enter all models only through their means over the four weeks before the current week, so that they precede the outcome; values of the current week are not used. The original analysis also used the current week's values. In the boosting models, as in the original analysis, precipitation is centred by its global mean for the calendar month over 2018–2022 (a single mean for all cells, not a local anomaly), and the other climate variables enter on their original scale. The boosting models also receive the month and season as features. A raw temperature means different things in different climates. In the latent models, the five climate variables (maximum and minimum temperature, precipitation, drought and $PM_{2.5}$) therefore enter as standardised anomalies with respect to a cell- and calendar-month-specific climatology estimated from the training years 2018–2022,

$$z_{st} = \bigl(x_{st} - \bar{x}_{s,\mathrm{m}(t)}\bigr) / \mathrm{sd}_{s,\mathrm{m}(t)},$$

where $\mathrm{m}(t)$ is the calendar month of week t. Precipitation is transformed to $\log(1+x)$ before the anomaly is computed. In dry seasons of arid cells the training-period standard deviation can be zero or close to zero, and any later rain would then give an extreme anomaly. We therefore floor $\mathrm{sd}_{s,\mathrm{m}}$ at the 10th percentile of the positive cell-month standard deviations of

the same variable, and set $z_{st} = 0$ where the climatological standard deviation is zero. In the AME region the floor applies, by construction, to 10% of the 164,628 cell-months for temperature and precipitation. The drought index is constant over the training years in 138,600 cell-months (84%), so its anomaly is zero there, and the floor applies to a further 1.3%. The drought anomaly is therefore informative only in a minority of cell-months. The $PM_{2.5}$ anomaly is worse. Its climatological standard deviation is undefined in 91,392 cell-months (56%), where $PM_{2.5}$ is missing, and the floor computed from the other cell-months is about 10^{-9}, so it does not prevent extreme anomalies. The $PM_{2.5}$ anomaly is missing in 55.5% of training rows, is then set to zero and has no missingness indicator. We do not interpret its coefficient. The anomaly coefficients are identified from variation over time within cells. They are therefore not confounded by the time-invariant spatial pattern of the climatology. They can still be confounded by regional shocks that vary over time, since the year effects are global. The cell's mean over the training years enters separately as a cell-level covariate. A five-year climatology is noisy, especially for precipitation, and a longer reference period would be preferable where available. The anomalies are therefore best read as departures from recent conditions.

Missing covariates.. Table 1 hides substantial missingness. In the repaired table over the AME region (4,976,634 cell-weeks), the IPC phase populations are missing in 62.7% of rows, $PM_{2.5}$ in 55.5%, and GDP in 3.6%, mainly in South Sudan, Yemen, Somaliland, Eritrea, Western Sahara and Syria. The other national indicators (EPI, liberal democracy index, greenhouse gas emissions, material and ecological footprint) are missing in 0.3% of rows, at most about 850 cell-weeks in any country. These shares are after the fallback described in Section 2.1, which fills some missing values with a neighbouring country's value. The climate means over the previous four weeks are missing in 1.1% of rows, and the event histories in 1.1%. Over all cell-weeks worldwide the IPC phases are missing in 89.6% of rows and precipitation in 20.8%. IPC analyses are carried out mainly where food crises are expected, and several of the countries without GDP data are among the most conflict-affected in the region. Missingness is therefore unlikely to be at random with respect to the outcome.

Apart from that fallback, we keep missing values as missing through the construction of the features. In particular, a cell-week without an IPC analysis has a missing IPC value, not a food-insecure population of zero. All covariates of the latent models other than the event histories are standardised with training-period means and standard deviations. Missing values are then set to zero, the training mean, and a missingness indicator enters the model for each covariate group: one for each national indicator and one for IPC (Section 3.1). The boosting models receive the missing values as such.

2.4. *Exploratory evidence of residual spatial dependence.* Before introducing the latent models, we check whether a flexible model of the covariates and recent history leaves spatial structure in its residuals. We take the corrected XGBoost model of Section 5.2, trained on all locations, and compute the standardised cell residuals of Section 3.10 over the test period 2023–2024. In the AME region, Moran's I of these residuals is 0.42 for conflict and 0.15 for unrest. The residual correlation between neighbouring cells is 0.45 within countries and 0.21 across borders for conflict, and 0.16 and 0.08 for unrest. Table S7 of the Supplementary Material gives the same diagnostics for every model, and the semivariogram is reported in Section 5.4. The test period contains the onsets of the wars in Sudan and Gaza, which no model trained on 2018–2022 can anticipate, so these residuals mix persistent structure with unforeseeable new events. The training period gives a check that does not depend on them. The corrected model is fitted to the training years, so its training residuals are in-sample, but the XGBoost offset cross-fitted by country (Section 3.8) predicts each training row from a

TABLE 1
Covariates, resolution and source.

Covariate	Resolution	Source
Minimum / maximum temperature	cell–week	CHIRTS
Heat index, wet-bulb globe temperature	cell–week	CHIRTS
Precipitation	cell–week	CHIRPS
$PM_{2.5}$	cell–week	CAMS
Drought index	cell–week	Global Drought Observatory
Events of each type in previous 4 weeks	cell–week	ACLED
Population	cell	GHSL
River length, lake area	cell	Natural Earth
Malaria risk	cell	Malaria Atlas Project
Biome	cell	RESOLVE Ecoregions
Tropical indicator ($\lvert\text{latitude}\rvert < 30°$; boosting only)	cell	derived
Food insecurity (IPC phases 1–5)	subnational–period	IPC
GDP per capita	country–year	World Bank
Access to electricity	country–year	World Bank
Environmental Performance Index	country–year	Yale EPI
Liberal democracy index	country–year	V-Dem
GHG emissions, material and ecological footprint p.c.	country–year	OWID, GFN

model that did not see its country. Over 2018–2022 its residuals have Moran's I of 0.64 for conflict and 0.20 for unrest. [TODO: Test-period diagnostics with Sudan and Gaza excluded.]

For the XGBoost offset cross-fitted by country (Section 3.8), the gap between the within-country and the across-border correlation is not by itself evidence of country-level effects. Neighbours in the same country are always predicted by the same fold model, which never saw that country, while neighbours across a border are usually predicted by different fold models. This design creates a country-level residual component. We therefore base the motivation for the latent effects on the corrected model, which has no such component, and report the cross-fitted offset's across-border pairs separately for countries in the same fold and in different folds (0.25 and 0.28 for conflict, on 264 and 1,776 pairs of cells). The two are similar, so in the test period the fold design does not appear to drive the across-border correlation of the offset. Over the training period they differ, 0.65 for pairs in the same fold and 0.41 for pairs in different folds for conflict, so there the fold design does contribute; for unrest they are 0.10 and 0.11. Residual structure that remains after the corrected model motivates the latent effects of Section 3.

3. Methods.

3.1. *A latent Gaussian model for event occurrence.* Let $\mathcal{S}$ be the set of n grid cells, $c(s)$ the country and $b(s)$ the biome of cell s, and $t = 1, \ldots, T$ the weeks. Conditionally on a set of latent effects, the outcomes are independent Bernoulli variables, $Y_{st} \mid \pi_{st} \sim \text{Bernoulli}(\pi_{st})$, with

$$\text{(3.1)} \quad \text{logit}\, \pi_{st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta} + \delta\, h_{st} + \sum_{k \in \mathcal{K}} z_{k,st}\, \beta_k(s) + \boldsymbol{z}_{c(s),t}^{\top}\boldsymbol{\alpha} + u_{c(s)} + w(s) + e_s + v(s,t).$$

Here h_{st} is the recent event history of Section 2.2, so δ measures state dependence (Section 3.4). The vector $\boldsymbol{x}_{st}$ holds the cell and cell-week covariates with fixed effects. In the application these are the other outcome's event history, population, malaria risk, the IPC phase 3+ population, river length and lake area, the climate anomalies and climatologies of

Section 2.3, biome effects and year effects. The variables $z_{k,st}$, $k \in \mathcal{K}$, are climate anomalies whose effects $\beta_k(s)$ vary in space (Section 3.6); in the application $\mathcal{K}$ contains maximum temperature only. The vector $\boldsymbol{z}_{ct}$ holds the national indicators with effects $\boldsymbol{\alpha}$ (Section 3.7). GDP per capita is one of them: it is national GDP divided by national population and enters as its logarithm. The country effects $u_c \sim \mathcal{N}(0, \sigma_u^2)$ are independent, w is a Matérn field on the sphere (Section 3.2), the cell effects $e_s \sim \mathcal{N}(0, \sigma_e^2)$ are independent, and v is a space-time field (Section 3.3). The four latent terms capture persistent *recorded* risk at four scales. Each mixes occurrence with reporting (Section 2.1): national institutions and reporting regimes (u), regional hotspots that cross borders (w), single cells such as a town, a mine or a border crossing (e), and slowly evolving episodes (v). The cell effect is needed because a field with $\nu = 1$ and a range of several cells cannot represent risk that differs between adjacent cells. Without it, such risk would be absorbed by w or by the history term δh_{st} (Section 3.4).

Missing covariate values are kept as missing through the construction of the features. In particular, a cell-week without an IPC assessment has a missing IPC value, not a food-insecure population of zero. After standardisation, missing values are set to the training-period mean, and a missingness indicator is added as a further fixed effect for each covariate group: one indicator for each national indicator and one for IPC. Without these indicators, the imputed value would be confounded with the fact of not being assessed, which is itself associated with conflict. The boosting models receive the missing values as such, since XGBoost handles them natively.

Model (3.1) is a latent Gaussian model (Rue, Martino and Chopin, 2009). Stacking all latent quantities into $\boldsymbol{\xi} = (\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}, \boldsymbol{u}, \boldsymbol{w}, \boldsymbol{e}, \boldsymbol{\eta}, \boldsymbol{v})$, the linear predictor is $\boldsymbol{\eta} = \boldsymbol{o} + \boldsymbol{A}\boldsymbol{\xi}$ for a sparse design matrix $\boldsymbol{A}$ and an optional offset $\boldsymbol{o}$, and $\boldsymbol{\xi} \mid \boldsymbol{\theta} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}(\boldsymbol{\theta})^{-1}\}$ with a sparse, block-diagonal precision whose blocks are listed in Table S1 of the Supplementary Material (Rue and Held, 2005). Fixed effects receive vague $\mathcal{N}(0, 10^4)$ priors and are integrated out together with the random effects.

3.2. *Matérn fields on the sphere via a finite-volume SPDE.* Following Lindgren, Rue and Lindström (2011), a Gaussian field w on $\mathbb{S}^2$ with Matérn-like covariance is defined as the stationary solution of

$$\tau\,(\kappa^2 - \Delta_{\mathbb{S}^2})\,w(s) = \mathcal{W}(s), \qquad s \in \mathbb{S}^2, \tag{3.2}$$

where $\Delta_{\mathbb{S}^2}$ is the Laplace–Beltrami operator and $\mathcal{W}$ is Gaussian white noise (see also Lindgren, Bolin and Rue, 2022). This is the $\alpha = 2$ case, which corresponds to Matérn smoothness $\nu = 1$ in the planar limit. The field then has practical range $\rho = \sqrt{8}/\kappa$ and marginal variance $\sigma^2 = 1/(4\pi\kappa^2\tau^2)$, provided ρ is small relative to the Earth's radius R. We parameterise the field by (ρ, σ), in kilometres and logit units.

The grid cells are natural control volumes, so instead of a triangulated mesh we discretise (3.2) by finite volumes on the longitude–latitude lattice itself (for finite-volume SPDE discretisations on regular grids, see Fuglstad et al., 2015). Integrating (3.2) over cell A_i and applying the divergence theorem with a two-point flux approximation gives

$$\tau\Big\{\kappa^2 a_i w_i + \sum_{j \sim i} \frac{\ell_{ij}}{d_{ij}}(w_i - w_j)\Big\} = \mathcal{W}(A_i) \sim \mathcal{N}(0, a_i),$$

where a_i is the area of cell i, $j \sim i$ ranges over the cells sharing an edge with i, ℓ_{ij} is the length of the shared edge and d_{ij} the great-circle distance between the centres. For a grid with spacings $\Delta\lambda, \Delta\varphi$ and centre latitude φ_i,

$$a_i = R^2 \Delta\lambda\,\{\sin(\varphi_i + \Delta\varphi/2) - \sin(\varphi_i - \Delta\varphi/2)\},$$
$$\ell_{ij}/d_{ij} = \Delta\varphi/(\cos\varphi_i\,\Delta\lambda) \quad \text{(east–west neighbours)},$$
$$\ell_{ij}/d_{ij} = \cos\varphi_{ij}\,\Delta\lambda/\Delta\varphi \quad \text{(north–south neighbours)},$$

with φ_{ij} the latitude of the shared edge. Writing $\boldsymbol{C} = \mathrm{diag}(a_i)$ and $\boldsymbol{G}$ for the weighted graph Laplacian with edge weights ℓ_{ij}/d_{ij}, we get $\tau \boldsymbol{K} \boldsymbol{w} = \boldsymbol{C}^{1/2} \boldsymbol{\epsilon}$ with $\boldsymbol{K} = \kappa^2 \boldsymbol{C} + \boldsymbol{G}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, so

(3.3)

$$\boldsymbol{w} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s^{-1}\}, \qquad \boldsymbol{Q}_s(\rho, \sigma) = \tau^2 \boldsymbol{K} \boldsymbol{C}^{-1} \boldsymbol{K}, \qquad \log|\boldsymbol{Q}_s| = 2n \log \tau + 2 \log|\boldsymbol{K}| - \textstyle\sum_i \log a_i.$$

$\boldsymbol{Q}_s$ has at most 13 non-zeros per row. The construction is the lumped-mass finite-volume analogue of the finite-element representation in Lindgren, Rue and Lindström (2011). Because it uses the physical cell geometry, it accounts for the shrinking east–west spacing towards the poles. It can also be restricted to land cells by deleting rows and columns, and it wraps around in longitude for global analyses. The continuum relations between (κ, τ) and (ρ, σ) hold on the lattice only when ρ spans several cells. A fitted range below two or three cells (100–150 km) indicates a field that the 0.5° lattice does not resolve; its $\hat{\sigma}$ is then not the marginal standard deviation, and we interpret such fields as cell-level effects.

REMARK 1. The discretisation imposes reflecting (Neumann) conditions on the boundary of the lattice, which roughly doubles the marginal variance of boundary cells (Section 4.2; see Khristenko et al., 2019, for the effect of boundary conditions on SPDE fields). For a land grid the boundaries are the coastlines. Extending the lattice into the sea removes the inflation but lets correlation leak across narrow seas, for example between Yemen and Somalia across the Gulf of Aden, which is substantively wrong for conflict risk. We therefore use the barrier model of Bakka et al. (2019): the SPDE is written as

$$u(s) - \nabla \cdot \left\{ \tfrac{r(s)^2}{8} \nabla u(s) \right\} = r(s) \sqrt{\pi/2}\, \sigma\, \mathcal{W}(s),$$

which reduces to (3.2) with $\kappa = \sqrt{8}/\rho$ when $r(s) \equiv \rho$. The lattice is extended by six rings of cells (3°) without data. Sea cells, identified from the Natural Earth country polygons, get range $r(s) = f\rho$. Land cells without data, such as those outside a regional analysis box, keep the range ρ, so that the field is not artificially decorrelated at the edge of the region. The finite-volume form is $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \boldsymbol{C} + (\rho^2/8)\boldsymbol{G}_m$, where $\boldsymbol{G}_m$ has edge weights $2m_i m_j/(m_i + m_j) \cdot \ell_{ij}/d_{ij}$, $m_i = r_i^2/\rho^2$ (the harmonic mean, the usual two-point flux choice for a discontinuous coefficient), and $\boldsymbol{D} = \mathrm{diag}\{r_i \sigma (\pi a_i/2)^{1/2}\}$, so that $\boldsymbol{Q}_s = \boldsymbol{K}\boldsymbol{D}^{-2}\boldsymbol{K}$. Table S2 of the Supplementary Material reports the marginal variance at the coast and the correlation across straits of two to six cells for several choices of f, at $\rho = 400$ km. At this range, $f = 0.1$, the value of Bakka et al. (2019), gives a barrier range of 40 km, less than one cell, and the coastline behaves almost like a reflecting boundary. We use $f = 0.5$. At $\rho = 400$ km it keeps the coastal variance within about 30% of nominal and the correlation across straits of four cells or more (about 220 km) below 0.04, while a strait of two cells still has a correlation of 0.16. Straits narrower than one cell are not resolved at this grid resolution. These figures depend on ρ, because the barrier range $f\rho$ and the buffer of six rings scale differently with it. The fitted ranges of the static fields in the application are mixed. In model (3.1) for Africa and the Middle East, $\hat{\rho}$ is 544 km for conflict and 465 km for unrest. In the regional fits of the state-dependence analysis it is 430–469 km for conflict and 333–335 km for unrest. In the joint model it is 453 km for w_C and 369 km for w_U. The fields of the spatially varying coefficients have ranges of 465 km (conflict) and 972 km (unrest). Below 400 km the barrier range $f\rho$ is shorter, and the coastline is closer to a reflecting boundary than in Table S2. For ranges of 800 km or more, such as that of the unrest coefficient field, $f\rho$ exceeds the buffer, so the coastal variance inflation returns and correlation leaks across the wider straits. The residual fields of the hybrid models (Section 3.8) have ranges of 263 km (conflict) and 65 km (unrest). The latter is not resolved by the lattice and is read as cell-level effects (Section 5.4). Table S2 is also computed for a sea strait inside a closed rectangle of

land, not for the geometry of the application, which has an open coast, a buffer of six rings and a reflecting outer edge. It shows the order of magnitude of the effects, not their size in the application. The stored real-data fits use the sea cells of the country polygons as barriers.

Priors.. We use penalised-complexity priors (Simpson et al., 2017; Fuglstad et al., 2019): $P(\rho < \rho_0) = 0.05$ and $P(\sigma > \sigma_0) = 0.01$, with $\rho_0 = 100$ km (about two grid cells) and $\sigma_0 = 1$. The PC prior of Fuglstad et al. (2019) is derived for Matérn fields on $\mathbb{R}^2$; we use it as an approximation on the sphere, which is adequate for ranges far below R. The country and cell effects have PC priors with $P(\sigma_u > 1) = 0.01$ and $P(\sigma_e > 1) = 0.01$.

3.3. *Space-time persistence.* The space-time term is a separable field that evolves as a first-order autoregression of Matérn innovations,

$$\boldsymbol{v}_\tau = \phi\, \boldsymbol{v}_{\tau-1} + \sqrt{1-\phi^2}\, \boldsymbol{\epsilon}_\tau, \qquad \boldsymbol{\epsilon}_\tau \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_v, \sigma_v)^{-1}\},$$

(Cameletti et al., 2013), so that $\boldsymbol{v}$ has precision $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$, where $\boldsymbol{Q}_{\mathrm{AR}}$ is the tridiagonal precision of a unit-variance AR(1) process. To control cost, τ indexes blocks of Δ weeks, and $v(s,t) = v_{s,\lceil t/\Delta \rceil}$. We use $\Delta = 1$ and $\Delta = 4$ in the simulations. In the application we use $\Delta = 8$ for the regional fits of the state-dependence analysis, and $\Delta = 13$ and $\Delta = 26$ for the space-time forecasting models and the joint model of Section 3.5. In the last two, v is also coarsened in space: it is defined on the 1° cells that contain observed 0.5° cells, and each 0.5° cell takes the value of its 1° cell. The field v is defined on the observed land cells only, with no buffer, so it cannot link cells across the sea. Its lattice has reflecting boundaries at the coast and at the edges of the analysis box. This departs from Remark 1, under which land cells outside a regional box keep the range ρ. Near these edges the prior variance of v is inflated, roughly doubled at an edge. The Central Sahel box of the state-dependence analysis is $22^\circ \times 11^\circ$, about 2400×1200 km. The fitted range of v there is 230 km for conflict and 485 km for unrest, so for unrest a large part of the box lies within one range of an edge, in particular in the north–south direction. Estimates of δ and κ conditional on v should be read with this caveat. In the state-dependence and joint models, the static field w uses the same lattice as elsewhere: the observed cells, a buffer of six rings and sea barriers with $f = 0.5$ (Remark 1). The space-time forecasting models contain no w. Their static part is the country effects and independent cell effects, so their predictor is the boosting offset plus $\beta_0 + u + e + v$. The prior on $\operatorname{atanh}\phi$ is $\mathcal{N}(0, 1.5^2)$. Non-separable alternatives exist (Lindgren et al., 2024) but are not needed for the questions asked here.

3.4. *State dependence versus persistent heterogeneity.* Recent event counts dominated the importance rankings of the original boosting analysis (Menicali, Marcantonio and Castruccio, 2026). Part of that dominance was an artefact of duplicated rows (Section 5.2), but after correction recent history remains the strongest single predictor. This has several explanations, which model (3.1) separates only partially:

1. *state dependence*: recorded events raise the probability of further recorded events in the same place, through retaliation, mobilisation or the arrival of armed groups;
2. *persistent heterogeneity*: some places are simply riskier (u, w, e or v is large there), and past events are informative only as a proxy for that latent risk (Heckman, 1981);
3. *reporting dynamics*: media attention and reporting networks themselves persist, so a place that has been reported on is more likely to be reported on again, whatever happens there (Weidmann, 2016).

All three produce strong autocorrelation in Y_{st} and a large predictive contribution of h_{st}. In the conflict literature the distinction appears as the question of whether apparent contagion

between neighbouring places reflects spillover or shared conditions (Buhaug and Gleditsch, 2008). Self-exciting point process models address it by decomposing intensity into a background and a triggering component (Mohler et al., 2011; Mohler, 2013; Porter and White, 2012; Clark and Dixon, 2018; Reinhart, 2018; Jun and Cook, 2024).

In model (3.1), δ is the effect of recent recorded history conditional on the latent persistence $u + w + e + v$, and it is identified only under the assumed latent structure. It is conditional on heterogeneity at the scales that are modelled. Persistent heterogeneity that none of the latent terms can represent loads onto δh_{st} and inflates $\hat{\delta}$. This is why the cell effect e is included: without it, time-invariant risk that varies between adjacent cells could only be represented by h_{st}. The space-time field v is resolved at steps of Δ weeks, and h_{st} covers the previous four weeks. When the two time scales are similar, separating δ from v relies on the AR(1) form, so we report δ for several Δ (Section 4.5). We therefore read $\hat{\delta}$ as conditional dependence of recorded events, not as a causal contagion effect. To limit the initial-conditions problem of dynamic random-effects models (Wooldridge, 2005), we drop weeks whose four-week history is incomplete (the first weeks of each cell and of each country's ACLED coverage). We also add each cell's history at its first usable week as a covariate, following Wooldridge (2005). This is a limited version of that device. A single four-week window is a noisy proxy for a cell's initial state, and its coefficient does not vary in time, whereas under the AR(1) form of v the dependence of later weeks on the initial state decays with time. The device therefore only partly removes the initial-conditions bias.

3.5. *A joint model of conflict and unrest.* Recent unrest is associated with later conflict in the same and in neighbouring places. This association has two readings. Unrest may escalate into armed conflict, for example through repression (Ritter and Conrad, 2016; Rød and Weidmann, 2023; Ryckman, 2020; Chenoweth, Perkoski and Kang, 2017). Alternatively, both outcomes may be driven by the same latent episodes, such as a period of political crisis in a region, or a period of intense media attention to it, without any effect of one on the other. Forecasting studies use unrest histories and their spatial lags as predictors of conflict (Rød, Hegre and Leis, 2025), and diffusion models add spatial lags of past events to the linear predictor (Racek, Thurner and Kauermann, 2026). Neither separates the two readings. We do so by modelling both outcomes jointly, with a space-time field that is common to them.

Let $j \in \{\mathrm{C}, \mathrm{U}\}$ index conflict and unrest, and let Y_{jst} be the indicator of a recorded event of type j in cell s and week t. The histories are $h_{jst} = \log(1 + \text{number of events of type } j \text{ in cell } s \text{ in weeks } t-4, \ldots, t-1)$, and $\bar{h}_{jst}$ is the mean of $h_{js't}$ over the rook neighbours s' of s, standardised on the training rows. Conditionally on the latent effects, the Y_{jst} are independent Bernoulli variables with

(3.4)

$$\operatorname{logit} \pi_{\mathrm{C}st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta}_{\mathrm{C}} + \delta_{\mathrm{C}}\, h_{\mathrm{C}st} + \kappa\, h_{\mathrm{U}st} + \delta_{\mathrm{C}}^{\mathrm{nb}}\, \bar{h}_{\mathrm{C}st} + \kappa^{\mathrm{nb}}\, \bar{h}_{\mathrm{U}st} + u_{\mathrm{C},c(s)} + e_{\mathrm{C}}(s) + w_{\mathrm{C}}(s) + v(s,t),$$

$$\operatorname{logit} \pi_{\mathrm{U}st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta}_{\mathrm{U}} + \delta_{\mathrm{U}}\, h_{\mathrm{U}st} + \kappa_{\mathrm{U}}\, h_{\mathrm{C}st} + \delta_{\mathrm{U}}^{\mathrm{nb}}\, \bar{h}_{\mathrm{U}st} + \kappa_{\mathrm{U}}^{\mathrm{nb}}\, \bar{h}_{\mathrm{C}st} + u_{\mathrm{U},c(s)} + e_{\mathrm{U}}(s) + w_{\mathrm{U}}(s) + \lambda\, v(s,t).$$

The vector $\boldsymbol{x}_{st}$ holds the covariates of model (3.1) that have fixed effects, including the initial-conditions term, the national indicators and the missingness indicators. All fixed effects are specific to the outcome. There are no spatially varying coefficients. For each outcome, $u_{j,c}$ are independent country effects, e_j independent cell effects and w_j a static Matérn field with sea barriers (Remark 1). The two sets of these terms are independent of each other, with their own hyperparameters. The space-time field v is the AR(1) $\otimes$ Matérn field of Section 3.3. It enters the conflict predictor with loading one and the unrest predictor with loading λ. The symbol κ here denotes the escalation coefficient. It is unrelated to the SPDE parameter of Section 3.2.

The copy construction.. A loading in the design matrix would make $\boldsymbol{A}$ depend on $\boldsymbol{\theta}$. We instead keep $\boldsymbol{A}$ fixed and put λ in the prior precision, as in the copy construction of R-INLA. Let $\boldsymbol{a}$ be the vector of the m values of v and introduce a second vector

$$\boldsymbol{b} = \lambda \boldsymbol{a} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \tau_c^{-1} \boldsymbol{I}), \qquad \tau_c = 10^4.$$

Conflict rows of the design load on $\boldsymbol{a}$ and unrest rows on $\boldsymbol{b}$. The standard deviation of $\boldsymbol{\varepsilon}$ is 0.01, small compared with that of v, so the unrest predictor receives λv up to a negligible perturbation. If $\boldsymbol{Q}_a = \boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$ is the precision of $\boldsymbol{a}$, the precision of $(\boldsymbol{a}, \boldsymbol{b})$ is

$$\boldsymbol{Q}_{ab} = \begin{pmatrix} \boldsymbol{Q}_a + \lambda^2 \tau_c \boldsymbol{I} & -\lambda \tau_c \boldsymbol{I} \\ -\lambda \tau_c \boldsymbol{I} & \tau_c \boldsymbol{I} \end{pmatrix} = \boldsymbol{R}_{ab}^\top \boldsymbol{R}_{ab}, \qquad \boldsymbol{R}_{ab} = \begin{pmatrix} \boldsymbol{R}_a & \boldsymbol{0} \\ -\lambda \tau_c^{1/2} \boldsymbol{I} & \tau_c^{1/2} \boldsymbol{I} \end{pmatrix},$$

with $\boldsymbol{R}_a^\top \boldsymbol{R}_a = \boldsymbol{Q}_a$, and $\log |\boldsymbol{Q}_{ab}| = \log |\boldsymbol{Q}_a| + m \log \tau_c$. The loading therefore does not enter the log-determinant, and the explicit factor $\boldsymbol{R}_{ab}$ is used for posterior sampling (Section 3.9). The sparsity pattern of $\boldsymbol{Q}_{ab}$ does not depend on λ. The loading has a $\mathcal{N}(0, 2^2)$ prior. The loading of conflict is fixed at one, which fixes the scale and the sign of v, so λ and its sign are identified. With $\lambda = 0$, the conflict equation of (3.4) is the univariate conflict model with its own space-time field, and the two outcomes are independent given their histories. The comparison of the joint fit with that univariate fit shows the effect of letting unrest inform v, under the assumption that there is no outcome-specific space-time variation (see *Identification and limits* below). In the application v uses $\Delta = 13$ weeks on the 1° lattice of Section 3.3.

The estimand.. The coefficient κ is the association between recent unrest in a cell and conflict in the same cell, conditional on the shared episode $v(s,t)$ and on the other terms of (3.4); κ^{nb} is the same for unrest in the neighbouring cells. We call κ escalation conditional on shared episodes. Like δ (Section 3.4), it is a conditional association of recorded events, not a causal effect.

Univariate models of conflict estimate a different quantity, even when they include their own space-time field. Suppose shared episodes drive both outcomes. Unrest in the previous four weeks was then generated in part by the episode at that time, and the episode persists, so $h_{\mathrm{U}st}$ is correlated with $v(s,t)$. A univariate conflict model treats its latent field as independent of the covariates. It estimates the field from conflict outcomes alone, and each latent value is informed by few Bernoulli observations, so the estimate is noisy and shrunk towards zero. The part of the episode that the field does not capture is still visible in $h_{\mathrm{U}st}$, which is an observed, if noisy, proxy of the episode. The coefficient of $h_{\mathrm{U}st}$ then absorbs it, and $\hat{\kappa}$ is biased upward. This is the random-effects exogeneity problem of Sections 3.4 and 3.7 in another form. In the joint model the unrest outcomes enter the likelihood of v. The episode is estimated from both outcomes, and the dependence of $h_{\mathrm{U}st}$ on it is part of the model, because $h_{\mathrm{U}st}$ is built from earlier values of a modelled outcome. Section 4.7 checks this argument by simulation. The simulation as reported has a single shared space-time field and no outcome-specific space-time field in the data-generating process, with steps of four weeks on a 0.5° lattice. It checks the upward bias of the univariate estimate, but not the bias in the opposite direction described below.

Identification and limits.. The shared field is identified from the joint movement of the two outcomes at its own resolution, beyond what the fixed effects and the outcome-specific static terms explain. The coefficient κ is identified from variation in four-week unrest histories within that resolution. Two consequences follow. First, escalation that unfolds over several months and over a region the size of the range of v cannot be told apart from a shared episode, and is attributed to v. The estimate $\hat{\kappa}$ is therefore the short-run, local part of escalation, and it can understate escalation at longer time scales. Second, shared episodes that are shorter than Δ weeks or finer than the lattice of v remain in κ. Third, episodes that affect unrest only are

also forced into v, because the model has no unrest-specific space-time field. Since v enters the conflict predictor with loading one, conditioning on v then removes part of the variation in $h_{\mathrm{U}st}$ through which escalation would act. This biases $\hat{\kappa}$ towards zero, even when escalation is real. The drop from the univariate to the joint estimate is therefore consistent both with the removal of confounding by shared episodes and with this attenuation. To separate the two, we fit an extended joint model with an additional unrest-specific space-time field of the same form as v, and we extend the simulation of Section 4.7 with scenarios that contain unrest-only episodes, with and without escalation [TODO: biv2 and S5 extension results].

The shared component mixes common shocks to occurrence with common shocks to reporting. Reporting rates vary in space (Weidmann, 2016), reporting of protest clusters in space and time through sequential attention (Hellmeier, Weidmann and Rød, 2018), and non-fatal events are under-reported selectively (Croicu and Eck, 2022). A period of attention to a place raises the recorded counts of both event types, and the joint model treats it as a shared episode. The two sources of co-movement could be separated with several independent sources of event data (Cook et al., 2017). With a single source they cannot. We therefore read κ as conditional on shared episodes of either kind, and we do not interpret v as occurrence or as reporting. The loading λ also mixes the two. It reflects how strongly unrest follows the shared episodes and how differently the two event types are reported.

Model (3.4) is a linear model of coregionalisation with a single space-time component and fixed loading one for conflict (Gelfand et al., 2004). Outcome-specific persistent variation enters through u_j, e_j and w_j, but there is no outcome-specific space-time field. Space-time variation of conflict that is not shared with unrest therefore also loads on v, and λ is estimated against it. Space-time variation of unrest that is not shared with conflict loads on v as well, with the attenuation of $\hat{\kappa}$ described above. A second space-time component, as in the extended model, relaxes this at the cost of a second latent block of the size of v. Multivariate Hawkes models of event data (Jun and Cook, 2024) and discrete-time Hawkes models of ACLED (Browning et al., 2026) explain co-occurrence by triggering, with a background rate that does not contain a shared time-varying latent component. In such models a shared episode can only appear as cross-excitation. Model (3.4) can be read as adding a shared stochastic background to cross-lagged terms. Spatio-temporal SPDE models of political violence (Python et al., 2019; Egbon et al., 2024) have so far modelled one outcome at a time.

3.6. *Spatially varying climate effects.* To assess whether climate effects differ across biomes and how much variation remains within them, the effect of anomaly k is decomposed as

$$\beta_k(s) = \gamma_{k,b(s)} + \eta_k(s), \qquad \boldsymbol{\eta}_k \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_k, \sigma_k)^{-1}\}, \tag{3.5}$$

a spatially varying coefficient model (Gelfand et al., 2003; Dambon, Sigrist and Furrer, 2021) whose mean is pooled by biome. The biome means $\gamma_{k,b}$ are fixed effects with the vague prior of Section 3.1. The field $\boldsymbol{\eta}_k$ has PC priors with $P(\rho_k < 100\,\mathrm{km}) = 0.05$ and $P(\sigma_k > 0.5) = 0.01$, and σ_k measures heterogeneity within biomes. The design of $\boldsymbol{\eta}_k$ is the sparse matrix with entry $z_{k,st}$ in row (s,t) and column s.

Three summaries of a biome-specific effect should be distinguished:

- *Fixed biome slopes.* A model with fixed biome-specific slopes and no $\boldsymbol{\eta}_k$ estimates an information-weighted average, $\sum_{s\in\mathcal{S}_b} \omega_s \beta_k(s) / \sum_{s\in\mathcal{S}_b} \omega_s$, with $\omega_s \approx \sum_t z_{k,st}^2 \pi_{st}(1-\pi_{st})$. This average puts most weight on the high-risk cells of the biome.
- *Biome means.* Model (3.5) targets the biome mean $\gamma_{k,b}$, the mean of the process that generated the biome's coefficients. This is a superpopulation quantity. Because $\boldsymbol{\eta}_k$ is not constrained to average zero within biomes and biomes are large contiguous regions, $\gamma_{k,b}$ is

separated from the biome average of $\boldsymbol{\eta}_k$ mainly when ρ_k is small relative to the size of the biome. When ρ_k is comparable to biome size, the separation rests partly on the prior. A nearly constant component of $\boldsymbol{\eta}_k$ is then confounded with a common shift of all $\gamma_{k,b}$. The intervals for $\gamma_{k,b}$ are conditional on $(\hat{\rho}_k, \hat{\sigma}_k)$ (Section 3.9), and in this situation the uncertainty in these hyperparameters may be the largest source of error. They should be read with this in mind.

- *Median SHAP value.* The summary used in the SHAP-based analysis of Menicali, Marcantonio and Castruccio (2026), the median SHAP value within a biome, differs from both. The following identities are on the logit scale of the model. They hold for interventional SHAP values when the features are independent. The path-dependent TreeSHAP of XGBoost (Lundberg et al., 2020), which we use, satisfies them only approximately, and features correlated with biome, such as distance from the equator, can absorb part of an interaction. For a feature with an additive effect $f_k(z)$, the SHAP value is $f_k(z) - \mathbb{E}f_k(Z_k)$, so the median within a biome is $\operatorname{median}_{(s,t):\, b(s)=b}\{f_k(z_{k,st}) - \mathbb{E}f_k(Z_k)\}$. For a linear effect $\beta_k z$ this is β_k times the median of $z_{k,st} - \mathbb{E}z_k$ in the biome. It reflects where the biome's covariate values lie relative to the global mean as much as the strength of the effect. When the effect is a biome-by-feature interaction, SHAP also splits the interaction between the feature and biome (Kumar et al., 2020; Aas, Jullum and Løland, 2021). A pure interaction $g(b)z$, with B independent of Z, contributes $\{g(b) + \mathbb{E}g(B)\}(z - \mathbb{E}z)/2$ to the SHAP value of z. Within biome b, the least-squares slope of the SHAP value of z on z therefore targets $\{g(b) + \mathbb{E}g(B)\}/2$, which is pulled towards the average effect across biomes. It equals $g(b)/2$ only when $\mathbb{E}g(B) = 0$. The SHAP value with the interaction credited fully to z, $\Phi_{zz} + 2\sum_{j \neq z} \Phi_{zj}$ with Φ the SHAP interaction values (Lundberg et al., 2020), equals $g(b)(z - \mathbb{E}z)$ in this case, so its slope within biome b targets $g(b)$ whatever the mean of g.

3.7. *National indicators and spatial confounding.* National indicators $\boldsymbol{z}_{ct}$, such as GDP per capita, the EPI or the ecological footprint, take one value per country and year. GDP per capita is computed at the national level (Section 3.1), so that it is constant within each country and year like the other indicators. We decompose each into a between-country component, its country mean over the training years $\bar{z}_c$, and a within-country component $z_{ct} - \bar{z}_c$, with separate coefficients (a Mundlak decomposition). Year effects absorb trends common to all countries. The between coefficients are identified from contrasts between the n_c countries only, so their effective sample size is of order n_c, not the number of cell-weeks. Modelling cell-weeks as conditionally independent, or bootstrapping spatial blocks that cut across countries, therefore produces intervals that are far too narrow. The within coefficients rest on the few years of change within each country. A missing value of an indicator is set to the training mean, and the indicator's missingness indicator is set to one (Section 3.1). The missingness indicator is defined per country and year. It varies mostly between countries, so its coefficient also rests on few contrasts. An indicator that is never missing in the training years has a constant missingness indicator there, which is dropped. This is the case for the EPI and liberal democracy. Their missing values in the test years are therefore set to the training mean without a flag (Section 5.6).

If the indicators are associated with unobserved country-level risk, so that $\mathbb{E}\{u_c + \bar{w}_c \mid \bar{z}_c\} \neq 0$ with $\bar{w}_c$ the country average of w, then the between coefficients are confounded. An indicator of environmental performance that tracks state capacity, or reporting capacity, is the leading example. This is the spatial confounding problem (Reich, Hodges and Zadnik, 2006; Hodges and Reich, 2010; Paciorek, 2010). Adding a spatial random effect changes the estimate of a spatially smooth covariate's coefficient, and in general neither the estimate with the random effect nor the one without it is unbiased for a causal effect (Khan and Calder,

2022; Zimmerman and Ver Hoef, 2022). Recent work has made this more precise. Generalised least squares with a spatial random effect can be consistent when the covariate has non-spatial variation (Gilbert, Ogburn and Datta, 2025). Confounding in the data-generating process should be distinguished from the behaviour of the analysis model (Khan and Berrett, 2026). The bias depends on how the covariate and the confounder are spread across spatial scales (Guan et al., 2023; Dupont, Marques and Kneib, 2026). A country-constant indicator has no within-country spatial variation, so these results give little reassurance here.

We report the following estimators and treat their disagreement as a measure of how fragile the national-level conclusions are:

1. model (3.1) without latent effects;
2. model (3.1) with $\boldsymbol{u}$ (the cell effects $\boldsymbol{e}$ enter only together with $\boldsymbol{w}$);
3. model (3.1) with $\boldsymbol{u}$, $\boldsymbol{w}$ and $\boldsymbol{e}$;
4. a *spatial+* estimator (Dupont, Wood and Augustin, 2022): model (3.1) with $\boldsymbol{u}$, $\boldsymbol{w}$ and $\boldsymbol{e}$, in which each between component is replaced by its residual after an area-weighted projection, over the observed land cells, on the m smoothest eigenvectors $\boldsymbol{\psi}$ of $\boldsymbol{G\psi} = \mu \boldsymbol{C\psi}$.

These eigenvectors are the lattice analogue of the thin-plate eigenvectors of Dupont, Wood and Augustin (2022). A country-constant indicator varies only by steps at borders. After the projection, the residual keeps these steps, minus a smooth function of the eigenvectors. The projection removes the part of the between component that is smooth on the scales of $\boldsymbol{w}$, so that it cannot be confounded with $\boldsymbol{w}$. It does nothing about confounding with $\boldsymbol{u}$. The remaining border steps are country-level jumps, which $\boldsymbol{u}$ can absorb. With $\boldsymbol{u}$ in the model, the between coefficient is therefore identified partly through the shrinkage of $\boldsymbol{u}$ under its exchangeable prior, which is independent of the indicator. This is the random-effects exogeneity assumption $\mathbb{E}(u_c \mid \bar{z}_c) = 0$, which we have just called implausible. The same holds for estimators (ii) and (iii): the between component $\bar{z}_c$ and u_c vary at the same country level, so the Mundlak decomposition does not protect the between coefficients. Their larger standard errors reflect the n_c contrasts, not robustness to confounding. The between conclusions under every estimator are conditional on this assumption. The projection also gives the residual some within-country variation that is a function of the smooth eigenvectors. We therefore treat (iv) as a sensitivity analysis and not as a preferred estimate.

The default m^* is the number of eigenvalues below $\kappa_w^2 = 8/\hat{\rho}_w^2$, the eigenvectors that vary on scales longer than the fitted range of w. We take $\hat{\rho}_w$ from estimator (iii): 544 km for conflict and 465 km for unrest. The eigenvalues are computed on the 13,719 observed land cells, up to a maximum of 400 eigenpairs. This gives $m^* = 138$ for conflict and $m^* = 176$ for unrest, both below the cap. By Weyl's law, for a region of area $|D|$ the count is approximately $|D|\kappa_w^2/(4\pi) = 2|D|/(\pi\hat{\rho}_w^2)$. With $|D| \approx 3.9 \times 10^7$ km^2 this gives about 84 and 114. The computed counts are larger, which is consistent with the positive boundary correction to Weyl's law under reflecting boundaries and the long coastline of the region. We report $m = 10$, $m = 20$ and m^*, that is $m \in \{10, 20, 138\}$ for conflict and $m \in \{10, 20, 176\}$ for unrest. The restricted spatial regression of Reich, Hodges and Zadnik (2006) is not used, because its intervals are known to be anticonservative (Hanks et al., 2015; Khan and Calder, 2022).

3.8. *Hybrid models: boosting offsets with latent effects.* To retain the flexibility of gradient boosting for the many weakly informative covariates, we also fit a semiparametric version of (3.1). Its fixed part is replaced by the logit $\hat{f}(\boldsymbol{x}_{st})$ of an XGBoost classifier (Chen and Guestrin, 2016), which enters as an offset:

$$\operatorname{logit} \pi_{st} = \hat{f}^{(-k(s))}(\boldsymbol{x}_{st}) + \beta_0 + u_{c(s)} + \omega(s), \tag{3.6}$$

where ω is either the Matérn field w or independent cell effects. The offset is *cross-fitted by country*. Countries are assigned at random to $K = 5$ folds. For each fold k, $\hat{f}^{(-k)}$ is trained on the 2018–2022 rows of the other folds, with its number of trees chosen by early stopping on 2022 after training on 2018–2021. It then provides the offset for every row of fold k, in both the training and test periods. The offset and the latent effects are fitted on the Africa and the Middle East box only. Trees that receive national indicators can identify countries, and through population, biome and distance from the equator, approximate locations. Cross-fitting by country ensures that the offset of a cell was learned without any data from its own country, so that u does not count that country's data twice. It does not exclude data from neighbouring countries. The offset of a cell near a border can be learned from the outcomes of cells across the border, and ω is estimated from the residuals of those same cells. Regional hotspots that cross borders can therefore enter both the offset and ω. The fitted ω is a residual field. Its range and standard deviation are not comparable with those of w in model (3.1). The offset enters with its coefficient fixed at one, so fold models whose logits are over- or under-dispersed are not rescaled; u absorbs shifts between folds but not differences in scale. Inference for u and ω is conditional on the offset, whose estimation uncertainty is not propagated, so their posterior uncertainty is understated. Cross-fitting is used here in the loose sense of sample splitting, not with an orthogonal score as in Chernozhukov et al. (2018).

The price of cross-fitting is that the offset is weaker than a boosting model trained on all locations. The cross-fitted XGBoost model never sees the country it predicts, while u and ω are estimated from that country's training data. A gain of the hybrid model over the cross-fitted offset alone is therefore not evidence that the latent effects add anything. The hybrid models are compared, on identical test rows, with the corrected boosting model trained on all locations, with the location-aware boosting baseline, with each other (ω as a field or as independent cell effects) and with the persistence baseline (Section 3.10). The offset uses the same rows, features and hyperparameters as the corrected boosting model (Section 5.2). It is trained without class weights, so it is on the log-odds scale.

Model (3.6) differs from GPBoost (Sigrist, 2022, 2023), which alternates boosting steps with latent Gaussian model updates: here the boosting fit is fixed before the latent model is fitted. This makes the model cheap enough for the global data and allows the same Laplace machinery to be used. Random forests with spatially correlated errors (Saha, Basu and Datta, 2023) are a further alternative.

The boosting analysis of Menicali, Marcantonio and Castruccio (2026) trained XGBoost with positives weighted by $\omega_+ = (n_0/n_1)^{1/2}$. For the unrestricted minimiser of the logistic loss, this shifts the fitted log-odds by $\log \omega_+$ (King and Zeng, 2001). We therefore prior-correct the predictions of the weighted models, $\hat{f} \leftarrow \hat{f} - \log \omega_+$, before probabilistic scoring. For boosted trees the correction is only approximate, because the weights also act on the leaf penalty, on the minimum leaf weight, which is defined on weighted Hessians, and on early stopping. The scores of the weighted models therefore include any remaining miscalibration. Class reweighting does not improve discrimination and miscalibrates probabilities (van den Goorbergh et al., 2022), and the corrected boosting model is trained without it (Section 5.2).

3.9. *Inference.* For fixed hyperparameters $\boldsymbol{\theta}$, the posterior mode $\boldsymbol{\xi}^* = \arg\max_{\boldsymbol{\xi}} \{\log p(\boldsymbol{y} \mid \boldsymbol{\xi}) - \frac{1}{2}\boldsymbol{\xi}^\top \boldsymbol{Q}\boldsymbol{\xi}\}$ is found by damped Newton iterations,

$$\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} + \ell\,(\boldsymbol{Q} + \boldsymbol{A}^\top \boldsymbol{W}\boldsymbol{A})^{-1}\{\boldsymbol{A}^\top(\boldsymbol{y} - \boldsymbol{\pi}) - \boldsymbol{Q}\boldsymbol{\xi}\}, \qquad \boldsymbol{W} = \operatorname{diag}\{\pi_{st}(1 - \pi_{st})\},$$

with step length ℓ halved until the objective does not decrease. If ℓ falls below 10^{-6}, the step is taken even if the objective has decreased. For the joint model (3.4) the rows of both outcomes

are stacked, so $\boldsymbol{y}$, $\boldsymbol{\pi}$ and $\boldsymbol{W}$ run over the cell-weeks of both. The Laplace approximation to the marginal likelihood is

(3.7)
$$\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) = \log p(\boldsymbol{y} \mid \boldsymbol{\xi}^*) - \tfrac{1}{2}\boldsymbol{\xi}^{*\top}\boldsymbol{Q}\boldsymbol{\xi}^* + \tfrac{1}{2}\log|\boldsymbol{Q}| - \tfrac{1}{2}\log|\boldsymbol{Q}+\boldsymbol{H}|, \qquad \boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}^* \boldsymbol{A}.$$

The hyperparameters are set to $\hat{\boldsymbol{\theta}} = \arg\max\{\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\}$ by L-BFGS-B on the internal scale: logarithms of ranges and standard deviations, $\operatorname{atanh}\phi$, and the loading λ itself. The internal parameters are bounded: standard deviations to $[0.005, 20]$, ranges to $[10, 2\times10^4]$ km, $\operatorname{atanh}\phi$ to $[-3, 3]$ and λ to $[-5, 5]$. The bounds keep the inner problem well conditioned; an estimate at a bound would indicate a poorly identified hyperparameter. Gradients are forward differences with step 10^{-4} on the internal scale, taken inwards at an upper bound. Each evaluation warm-starts Newton at the previous mode. Given $\hat{\boldsymbol{\theta}}$, the latent posterior is approximated by $\mathcal{N}\{\boldsymbol{\xi}^*, (\boldsymbol{Q}+\boldsymbol{H})^{-1}\}$. Samples reuse the factorisation of $\boldsymbol{Q}+\boldsymbol{H}$ already computed at the mode, through the perturbation identity

$$\boldsymbol{\xi} = \boldsymbol{\xi}^* + (\boldsymbol{Q}+\boldsymbol{H})^{-1}\big(\boldsymbol{R}^\top \boldsymbol{\epsilon}_1 + \boldsymbol{A}^\top \boldsymbol{W}^{*1/2}\boldsymbol{\epsilon}_2\big),$$

where $\boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{Q}$ is block-wise explicit, for instance $\boldsymbol{R} = \boldsymbol{D}^{-1}\boldsymbol{K}$ for a Matérn block and a Kronecker product of a bidiagonal AR(1) factor with that matrix for $\boldsymbol{v}$, and $\boldsymbol{R}_{ab}$ of Section 3.5 for the shared field of the joint model. This is an empirical Bayes version of INLA (Rue, Martino and Chopin, 2009) and is similar in spirit to TMB (Kristensen et al., 2016). Laplace approximations are generally accurate for binary GLMMs but can understate uncertainty when there are few observations per latent variable and events are rare (Fong, Rue and Wakefield, 2010; Ferkingstad and Rue, 2015). The intervals also ignore the uncertainty in $\hat{\boldsymbol{\theta}}$. Section 4.2 therefore checks coverage at the event prevalence of the data, and Section 4.5 checks it for the space-time model, where each latent value is informed by only Δ Bernoulli observations. When a space-time field is fitted and the data contain one, the plug-in standard errors are too small. In S3 the ratio of the empirical standard deviation of $\hat{\delta}$ to its mean standard error is between 1.19 and 1.66, and the coverage of nominal 95% intervals is between 0.80 and 0.95. In S5 without escalation the ratio for $\hat{\kappa}$ is 1.20 (Section 4.7). Intervals from the space-time and joint models in the application are therefore likely too narrow, and an estimate whose nominal interval only just excludes zero should not be read as clearly different from zero.

Computation.. All prior precisions are sparse. Every Newton step and every log-determinant requires a factorisation of $\boldsymbol{Q}+\boldsymbol{H}$, whose sparsity pattern does not depend on $\boldsymbol{\theta}$. The design $\boldsymbol{A}$ has a few dense columns, those of the fixed effects, and many sparse ones. Writing $\boldsymbol{A} = (\boldsymbol{X}_d, \boldsymbol{S})$, we store $\boldsymbol{X}_d$ as a dense array and assemble $\boldsymbol{H} = \boldsymbol{A}^\top\boldsymbol{W}\boldsymbol{A}$ by blocks. The blocks $\boldsymbol{X}_d^\top\boldsymbol{W}\boldsymbol{X}_d$ and $\boldsymbol{S}^\top\boldsymbol{W}\boldsymbol{X}_d$ are accumulated over chunks of 250,000 rows by dense matrix products (BLAS), which bounds the memory used by each process. Only $\boldsymbol{S}^\top\boldsymbol{W}\boldsymbol{S}$ is computed as a sparse product. The dense blocks are stored with their explicit zeros, so the sparsity pattern of $\boldsymbol{Q}+\boldsymbol{H}$ is the same at every evaluation.

Two factorisations are implemented. The first is the supernodal Cholesky factorisation of CHOLMOD, through scikit-sparse. Its symbolic analysis, which includes the fill-reducing ordering, is computed once per model and reused for every numerical factorisation. The second, used when CHOLMOD is not available, is a sparse LU factorisation in symmetric mode (SuperLU). For matrices of dimension above 2,000 its fill-reducing ordering is a nested dissection computed once by METIS (Karypis and Kumar, 1998). If METIS is not available, the code falls back to multiple minimum degree on $\boldsymbol{M}^\top + \boldsymbol{M}$, with $\boldsymbol{M} = \boldsymbol{Q}+\boldsymbol{H}$. For the space-time model the nested-dissection ordering reduced factorisation time by a factor of 7 to 14 compared with that fallback. An LU factorisation stores both triangular factors, so it needs about twice the memory of a Cholesky factor of the same matrix. scikit-sparse was not

installed on the computing cluster, so the fits run there used SuperLU [TODO: confirm the backend of the Africa and the Middle East joint fit].

The n_θ perturbed evaluations of a finite-difference gradient are independent. They run in parallel in forked worker processes, which share the data, the design and the symbolic analysis without copying them and start Newton from the current mode. The joint model (3.4) has $n_\theta = 12$ hyperparameters: two for the country effects, two for the cell effects, four for the static fields and four for the shared field. Newton iterations stop when the relative change of the objective is below 10^{-8} and the largest step is below 10^{-5}, or after 50 steps. An inner solve that stops at 50 steps without converging gives a Laplace value that is meaningless and often spuriously high. Such evaluations are rejected, as are those at which the factorisation fails: the objective is set to a very large value. In a parallel gradient, a component whose perturbed evaluation is rejected is set to zero. If the evaluation at the current point is itself rejected, the components are not set to zero and take very large values. In the serial path, the finite differences of L-BFGS-B are taken on the large value directly. Rejected evaluations can make the optimiser stop at a poor point, so they should be recorded for each fit [TODO: report the number of rejected evaluations for the real-data fits].

Since $\boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W} \boldsymbol{A}$ aggregates over weeks, the cost of the spatial and SVC blocks is driven by the number of cells, not by the number of cell-weeks. A global 0.5° land grid gives about 6×10^4 latent values per spatial field. The space-time block has nT_v latent values, and twice as many in the joint model, which is why it is fitted on regional subsets or on a coarser lattice. The cost of forming $\boldsymbol{H}$ grows with the number of cell-weeks and the number of fixed effects. For the joint model both are doubled, since each outcome has its own rows and its own fixed effects.

Prediction.. For a new cell-week with design row $\boldsymbol{a}$ and offset o, the posterior predictive probability is $\mathbb{E}\{\operatorname{expit}(o + \boldsymbol{a}^\top \boldsymbol{\xi})\} \approx \operatorname{expit}\{m/\sqrt{1 + \pi s^2/8}\}$, where m and s^2 are the posterior mean and variance of $o + \boldsymbol{a}^\top \boldsymbol{\xi}$; s^2 is estimated from posterior samples. Test years beyond the training period take the year effect of the last training year.

3.10. *Diagnostics and evaluation.*

Residual diagnostics.. Binary residuals are uninformative one at a time, so we aggregate over the test weeks within cells. The standardised cell residual is

$$r_s = \sum_t (y_{st} - \hat{\pi}_{st}) \Big/ \Big\{ \sum_t \hat{\pi}_{st}(1 - \hat{\pi}_{st}) \Big\}^{1/2},$$

with the sums over the test period 2023–2024. All residual diagnostics reported in the paper are computed in this way for every model, including the cross-fitted offset, on the same test rows. Residuals of in-sample latent-model fits are shrunk and spatially dependent through the smoother, and we do not use training-period residuals as evidence.

Even for a correct model, the test residuals are neither standard normal nor independent across cells. The predictions $\hat{\pi}_{st}$ carry the estimation error of the training period, for example in $\hat{w}(s)$. A logit error ϵ_s contributes about $\epsilon_s \{\sum_t \pi_{st}(1 - \pi_{st})\}^{1/2}$ to r_s, which grows with the number of test weeks, and the errors of a smoothed field are positively correlated between neighbours. The diagnostics below are therefore used to compare models on the same rows, not as tests against a null of independence. We report:

- Moran's I (Moran, 1950) of $\{r_s\}$ for rook adjacency. We also give a permutation p-value from $B = 999$ permutations, which is bounded below by $1/(B+1)$. Its null hypothesis of exchangeable residuals does not hold even for a correct model, so it is descriptive only;
- the empirical semivariogram of $\{r_s\}$ against great-circle distance;
- the correlation of r_s between neighbouring cells in the same country and across a border.

A larger within-country correlation can point to effects at the country level, of occurrence or of reporting. For the cross-fitted offset it also has a mechanical source. Neighbours in the same country are always predicted by the same fold model, while neighbours across a border are usually predicted by different fold models, whose differences create a country-level residual component. For this model we therefore also split the across-border pairs into pairs whose two countries are in the same fold and pairs in different folds.

Out-of-sample evaluation.. Models are trained on 2018–2022 and evaluated on 2023–2024, matching the original analysis. All models are compared on identical test rows. We report the logarithmic score and the Brier score, both strictly proper (Gneiting and Raftery, 2007), and the area under the precision–recall curve, computed as average precision (Boyd, Eng and Page, 2013). The latter is more informative than the ROC curve at the low prevalence of the data (Saito and Rehmsmeier, 2015). Paired differences in log score and Brier score are reported with two standard errors: one clustered by cell and one clustered by country, with the number of clusters. The cell-clustered errors allow for dependence over time within a cell but not between cells. The latent fields and the national indicators induce dependence at regional and national scales (Section 3.7), so the country-clustered errors are the more reliable of the two, although they rest on few clusters. Since prediction in the same places is dominated by persistence (Mueller and Rauh, 2022), every model is compared with a persistence baseline that ranks cell-weeks by the number of events in the previous four weeks. We also report all scores on *onset* rows: test cell-weeks in cells that had no event of that outcome in the previous 52 weeks. On these rows recent history carries no information, so they show how well a model anticipates events in places that have been quiet.

We also compare every model with a *location-aware* boosting baseline that adds each cell's training-period event rate as a feature. For the fit used to choose the number of trees by early stopping on 2022, this rate is computed from 2018–2021 only, so it uses no 2022 outcomes. For the final fit on 2018–2022 it is computed leaving out the row's own year, and for the test rows it is the rate over 2018–2022. This baseline is the boosting counterpart of a cell intercept, and the latent fields have to improve on it to be worth their cost.

4. Simulation studies. The simulations address six questions that the application raises but cannot settle on its own:

(S0) whether the finite-volume SPDE and the empirical Bayes Laplace approximation recover the field and give calibrated intervals, including at the event prevalence of the data;
(S1) whether boosting leaves residual spatial structure, and whether modelling it improves forecasts for the same places beyond a location-aware boosting baseline;
(S2) how national indicators behave under spatial confounding;
(S3) whether state dependence can be separated from persistent heterogeneity;
(S4) whether biome-specific climate effects are recovered better by spatially varying coefficients than by SHAP summaries;
(S5) whether an association between recent unrest and conflict reflects escalation or shared latent episodes, and whether a joint model of the two outcomes separates them.

All code is in the supplementary material.

4.1. *Data-generating process.* The simulator mimics the structure of the ACLED grid. The domain is a 40×30 block of $0.5°$ cells between $10°$E and $30°$E and between $5°$S and $10°$N (1,200 cells of about 55 km), with 104 training weeks, 52 test weeks and 12 burn-in weeks. Countries are the Voronoi cells of randomly placed seeds (14 countries; 25 in S2). Four biomes are latitude bands with boundaries perturbed by a smooth random field, ordered from "cold" (biome 1) to "tropical" (biome 4). Population is a standardised Matérn field

(range 250 km) plus white noise. Temperature and precipitation anomalies are standardised AR(1) processes in time (coefficients 0.6 and 0.3) of Matérn fields with ranges of 800 and 500 km. A raw temperature is also generated, as a climatology $30 - 1.2(\varphi - \varphi_{\min})$ in latitude φ, which rises from biome 1 to biome 4, plus a seasonal cycle $(1 + 0.15|\varphi|)\cos(2\pi t/52)$ and 2.5 times the anomaly. The climatology and the seasonal cycle have no effect in the truth. Three national covariates Z_0, Z_1, Z_2 are standard normal across countries. Z_1 and Z_2 have no effect but, like the national indicators in the data, allow trees to identify countries. Outcomes are generated sequentially from

(4.1)
$$\operatorname{logit}\pi_{st} = \beta_0 + \beta_{\text{pop}}\text{pop}_s + \delta\log\Big(1+\textstyle\sum_{l=1}^{4} y_{s,t-l}\Big) + \beta(s)\,\text{temp}_{st} + \beta_{\text{prec}}\text{prec}_{st} + \alpha Z_{0,c(s)} + u_{c(s)} + w(s) + v(s,t),$$

with $\beta(s) = \gamma_{b(s)} + \eta(s)$ and $u_c \sim \mathcal{N}(0, 0.3^2)$. To avoid generating data from the model used for fitting, w, η and the innovations of v are drawn from Gaussian processes with the continuous Matérn covariance ($\nu = 1$) in great-circle distance, evaluated at the cell centres. They are not drawn from the lattice GMRF of Section 3.2. Unless stated otherwise, $\beta_0 = -4$, $\beta_{\text{pop}} = 0.5$, $\delta = 0.6$, $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$, $\eta \equiv 0$, $\beta_{\text{prec}} = -0.15$, $\alpha = 0$, $\sigma_w = 1$, $\rho_w = 400$ km and $v \equiv 0$. The resulting prevalence of event weeks is about 4%. Spatial confounding (S2) is induced by setting $Z_{0,c} = \varrho\,\tilde{\ell}_c + (1 - \varrho^2)^{1/2}\epsilon_c$, where $\tilde{\ell}_c$ is the standardised country average of $u_c + w(s)$ and $\epsilon_c \sim \mathcal{N}(0, 1)$.

The default parameters are not calibrated to the application. The hybrid model fitted to the conflict outcome in the AME region (Africa, the Middle East and adjacent regions; Section 5.4) has a field standard deviation of 1.48, a range of 263 km and a country standard deviation of 0.76, and there persistence is about as accurate as cross-fitted boosting (AUC-PR 0.45 and 0.46). S1 is therefore repeated with $\sigma_w = 1.5$, $\rho_w = 260$ km and $\sigma_u = 0.8$, other parameters at their defaults. All covariate effects in (4.1) are linear on the logit scale, and the only interaction is between biome and temperature. This setting does not reward the flexibility of trees, so the comparisons between boosting and latent models below hold for this data-generating process and need not carry over to the application.

With R replicates, a proportion p (coverage, rejection rate) has Monte Carlo standard error $\{p(1-p)/R\}^{1/2}$, at most 0.09 for $R = 30$ and 0.11 for $R = 20$, and a mean estimate has Monte Carlo standard error equal to its empirical SD divided by $R^{1/2}$. We give these standard errors where a comparison depends on them.

Boosting.. Two versions of XGBoost are used, both with the hyperparameters of the original analysis (Menicali, Marcantonio and Castruccio, 2026): depth 5, learning rate 0.05, row and column subsampling 0.8, minimum child weight 30 and L_2 penalty 5. The *original-like* version follows the original procedure. Positives are weighted by $(n_0/n_1)^{1/2}$. The number of trees is chosen by early stopping on the last 26 training weeks, after which the model is refitted on all training weeks. Its log-odds are prior-corrected before scoring (Section 3.8). The *unweighted* version uses 300 trees without class weights. The minimum child weight of 30 was chosen for millions of rows. At a prevalence of about 4% each row contributes a Hessian of about 0.04, so a leaf needs several hundred rows, which restricts the trees more on the simulated training sets (about 125,000 rows, or 37,000 for a 30% subsample) than on the data. The features are population, distance from the equator, biome, the two anomalies, Z_0–Z_2 and the four-week event count. As in the original analysis, they include no coordinates.

4.2. *S0: accuracy of the field representation and of the Laplace engine.* We draw w on the 40×30 domain from the continuous Matérn process with $\rho_w = 400$ km and $\sigma_w = 1$, and Bernoulli outcomes from $\operatorname{logit}\pi_{st} = \beta_0 + 0.5\,x_{st} + w(s)$ with $x_{st} \sim \mathcal{N}(0, 1)$, in two settings with 50 replicates each. In the *common* setting, $\beta_0 = -2$ with 40 weeks (prevalence about 16%). In the *rare* setting, $\beta_0 = -4.5$ with 104 weeks (prevalence about 2%, as in the data).

TABLE 2

S0: accuracy of the finite-volume SPDE field and of the empirical Bayes Laplace approximation, with fields drawn from the continuous Matérn process (50 replicates per setting). Mean over replicates, with the SD across replicates in parentheses, except for the coverage of b_1, where it is the Monte Carlo standard error.

Quantity	common ($b_0 = -2$, 50 reps)	rare ($b_0 = -4.5$, 50 reps)
Prevalence of event weeks	0.161 (0.030)	0.019 (0.005)
$\hat{\rho}_w$ (km) [truth 400]	420 (40)	409 (51)
$\hat{\sigma}_w$ [truth 1]	0.884 (0.064)	0.875 (0.075)
$\hat{b}_1$ [truth 0.5]	0.494 (0.016)	0.502 (0.021)
Coverage of 95% interval for b_1	0.88 (0.05)	0.96 (0.03)
Coverage of 95% intervals for $b_0 + w(s)$ (share of cells)	0.948 (0.009)	0.943 (0.017)
$\mathrm{cor}(\hat{w}, w)$	0.954 (0.008)	0.914 (0.018)
Fit time (s)	2.4 (0.7)	3.4 (0.8)
Marginal variance of w, interior cells [nominal 1]	1.07	–
Marginal variance of w, edge cells	2.02	–

We also report the marginal variance of the lattice field itself, from 2,000 exact draws, in interior cells and at the edge of the domain. Table 2 summarises the results.

The lattice field has marginal variance 1.07 in cells at least eight cells (about 440 km) from the edge of the domain, against a nominal 1, and 2.02 at the edge. This is the doubling induced by the reflecting boundary that motivates the barrier model of Remark 1. Although the data are not generated from the lattice model, the range is recovered with little bias in both settings ($\hat{\rho}_w = 420$ and 409 km against 400). The standard deviation is underestimated by about 12% ($\hat{\sigma}_w = 0.88$ against 1). The bias is the same at prevalences of 16% and 2%, so it is not mainly due to the limited information in rare binary outcomes. A plausible cause is the mismatch between the lattice model and the continuous process, including the inflated variance near the boundary, since only about 28% of the cells are at least eight cells from the edge. We have not tested this explanation. The posterior mean of w has correlation 0.95 with the truth in the common setting and 0.91 in the rare one. Plug-in 95% intervals cover b_1 in 88% of replicates in the common setting and 96% in the rare one (Monte Carlo SE about 4.6% and 2.8%). They cover $\beta_0 + w(s)$ for 94.8% and 94.3% of cells. Ignoring the uncertainty in $\boldsymbol{\theta}$ therefore costs little here, including at the prevalence of the data. This holds for the coefficient of a covariate that is independent in space and time and independent of w, which is the easiest case. S0 does not check spatially smooth covariates that are correlated with the field, and S3 shows that the plug-in standard error of δ is too small when a space-time field is fitted. A fit takes 2–4 seconds on one core.

4.3. *S1: residual structure after boosting.* Data are generated from (4.1) with the default parameters. All models are trained on the first 104 weeks and evaluated on the following 52 weeks at the same cells, as in the original analysis. We compare the true probabilities (oracle); the original-like XGBoost; the same model with each cell's training event rate as an additional feature, computed leaving out the row's own 52-week period (the location-aware baseline of Section 3.10); the unweighted XGBoost cross-fitted by country; the hybrid model (3.6) with independent cell effects and with the Matérn field; the parametric model (3.1) with country effects and field; and the persistence baseline. The parametric model has a single temperature slope, so it omits the biome-specific slopes of the truth and is not correctly specified. Table 3 reports out-of-sample scores over 20 replicates. Table S3 of the Supplementary Material reports Moran's I of the training-period cell residuals of the two XGBoost models and the recovery of the field. Figure 2 shows one replicate.

Three findings stand out. First, boosting leaves strong spatial structure in its residuals. Moran's I of the training-period cell residuals is 0.36 (SD 0.05 across replicates) for the original-like XGBoost, which is trained on the same locations, and 0.64 for the version cross-fitted by country. Under independence it is about -0.001. The trees recover part of the field through country-identifying national covariates, so in-sample residuals understate the spatial structure. The cross-fitted offset alone predicts poorly (AUC-PR 0.10 against 0.17), as it must, since it ignores everything that is specific to a country.

Second, the field estimated on top of the cross-fitted offset recovers the truth (correlation between $\hat{w}$ and w 0.85), with $\hat{\rho}_w = 409$ km against 400 and $\hat{\sigma}_w = 0.83$ against 1.

Third, modelling the latent structure improves forecasts for the same places, but against a boosting baseline that is given each cell's own history the gain is small, and it reverses in the calibrated setting below. Differences below are paired by replicate, with standard errors over the 20 replicates.

- *Against the original-like XGBoost.* The hybrid model with the field improves the log score in all 20 replicates (mean difference -0.0027, SE 0.0002) and AUC-PR in all 20 ($+0.011$, SE 0.001).
- *Field against independent cell effects.* The field beats independent cell effects in every replicate (log score -0.0013, SE 0.0001; AUC-PR $+0.003$, SE 0.0003): smoothing across neighbouring cells helps.
- *Against the location-aware baseline.* The location-aware boosting baseline is a strong competitor. It improves on the original-like XGBoost about as much as the hybrid model does. Against it, the hybrid model with the field has a better log score in 16 of 20 replicates (mean difference -0.0011, SE 0.0004) and a better AUC-ROC in 18 of 20 ($+0.008$, SE 0.001). Its Brier score is the same (difference $+0.00002$, SE 0.00006; better in 9 of 20) and its AUC-PR is similar (-0.001, SE 0.001; better in 6 of 20). This baseline uses each cell's unsmoothed event rate, computed from 52 weeks for training rows and from 104 weeks for test rows, so the feature is noisier in training than at test. It does not smooth across neighbouring cells, and we have not compared the field with a boosting model that does, such as one with neighbour rates or a Gaussian process component. The log-score gain closes about a quarter of the gap between the baseline and the true probabilities (0.0042). The hybrid model with independent cell effects is worse than the baseline: its log score is similar ($+0.0002$, SE 0.0004; better in 6 of 20) and its AUC-PR is lower (-0.004, SE 0.001; lower in 17 of 20).
- *Parametric model.* Although it omits the biome-specific temperature slopes, the parametric model beats the baseline in log score in every replicate (-0.0020, SE 0.0004) and in AUC-PR in 19 of 20 ($+0.004$, SE 0.001). It closes about half of the log-score gap to the true probabilities.

Once boosting is given each cell's own history, the gain from the latent field is therefore small. It appears in the log score and in AUC-ROC, but not in the Brier score or in AUC-PR, so these results do not show whether it comes from calibration or from ranking. The persistence baseline reaches an AUC-PR of 0.095, against 0.20 for the true probabilities. In the data, persistence is much stronger, because some cells record events almost every week. S1 does not reproduce that regime, which is one reason to repeat it with the calibrated parameters. With those parameters the prevalence of event weeks in the test period is 6.9% (SD 2.8% across replicates), and persistence reaches an AUC-PR of 0.30 against 0.47 for the true probabilities. The posterior mean of the field is about as close to the truth as before (correlation between $\hat{w}$ and w 0.84), but the hyperparameters are more biased: $\hat{\sigma}_w = 1.16$ against 1.5 and $\hat{\rho}_w = 299$ km against 260. The comparison with the location-aware baseline changes, however (paired differences, SE over 20 replicates):

TABLE 3

S1: out-of-sample scores on the 52 test weeks (mean over 20 replicates, Monte Carlo standard error in parentheses). Lower is better for the log and Brier scores. The last two columns give the number of replicates in which the model has a better log score than (a) the original-like XGBoost and (b) the location-aware XGBoost with the cell training rate.

Model	Log score	Brier	AUC-PR	AUC-ROC	< (a)	<
True probabilities	0.1343 (0.0070)	0.0334 (0.0022)	0.200 (0.015)	0.805 (0.004)	20/20	20
XGBoost as in the original analysis	0.1401 (0.0072)	0.0341 (0.0023)	0.172 (0.015)	0.767 (0.006)	–	2/
XGBoost + cell training rate	0.1385 (0.0070)	0.0338 (0.0022)	0.185 (0.015)	0.781 (0.005)	18/20	
XGBoost, cross-fitted by country	0.1537 (0.0085)	0.0360 (0.0026)	0.101 (0.011)	0.670 (0.008)	0/20	0/
Cross-fitted XGBoost + u + iid cell effects	0.1387 (0.0071)	0.0339 (0.0023)	0.181 (0.015)	0.778 (0.005)	19/20	6/
Cross-fitted XGBoost + $u + w$	0.1374 (0.0071)	0.0339 (0.0023)	0.184 (0.015)	0.789 (0.005)	20/20	16
Parametric model with $u + w$	0.1365 (0.0071)	0.0336 (0.0022)	0.189 (0.015)	0.792 (0.005)	20/20	20
Persistence baseline	–	–	0.095 (0.010)	–	–	

- the hybrid model with the field is now worse than the baseline: log score $+0.0011$ (SE 0.0008; better in 5 of 20), Brier score $+0.0006$ (SE 0.0002), AUC-PR -0.013 (SE 0.001; lower in all 20) and AUC-ROC -0.0013 (SE 0.0005). With independent cell effects it is worse again (log score $+0.0021$, SE 0.0008; AUC-PR -0.014, SE 0.001). The offset cross-fitted by country is much weaker here than the baseline (AUC-PR 0.26 against 0.45), and the latent effects do not make up the difference;
- the parametric model still beats the baseline: log score -0.0026 (SE 0.0007; better in all 20), Brier score -0.0005 (SE 0.0002; 17 of 20), AUC-PR $+0.003$ (SE 0.001; 15 of 20) and AUC-ROC $+0.004$ (SE 0.0005; all 20). It closes about 40% of the log-score gap between the baseline and the true probabilities (0.0064).

The gain from a latent field on top of boosting therefore depends on the setting, and with a stronger field and stronger country effects it disappears. The model that benefits in both settings is the parametric one, whose functional form matches the truth up to the omitted biome slopes.

4.4. *S2: national indicators under spatial confounding.* Three scenarios with 25 countries and 30 replicates each are considered: no confounding with $\alpha = 0$ ($\varrho = 0$); confounding with $\alpha = 0$ ($\varrho = 0.8$); and confounding with a true effect $\alpha = -0.3$. We compare estimators (i)–(iv) of Section 3.7, with spatial+ using $m = 10$ eigenvectors, and the XGBoost summary of the original analysis. For the latter, the original-like XGBoost is fitted to a fixed 30% subsample of the training data, as in the original procedure. The effect of Z_0 is summarised by the least-squares slope of its SHAP values on Z_0. Its intervals come from 40 replicates of a spatial block bootstrap with 2.5° blocks (48 blocks on the domain), each refitted with the same number of trees and class weight. The reported estimate is the median over bootstrap replicates, as in the original analysis. A single fit to the same subsample gives nearly the same estimates (means -0.01, 0.34 and 0.15 in the three scenarios below, against bootstrap medians of -0.01, 0.33 and 0.14). The estimate from a single fit to all training rows is not closer to the truth: its means are -0.01, 0.40 and 0.17 (empirical SD 0.08–0.10). The bias under confounding is therefore not an artefact of the subsample or of the bootstrap.

Figure 3 summarises the results (Table S4 of the Supplementary Material gives the numbers). Without confounding, all estimators are centred on zero, but intervals that ignore the country-level structure of the covariate are badly anticonservative. Logistic regression reports a standard error of 0.016 against an empirical SD of 0.107. It covers in 23% of replicates and declares a non-existent effect significant in 77%. Country effects alone restore nominal coverage. The block-bootstrap summary of the original analysis is closer to nominal than logistic

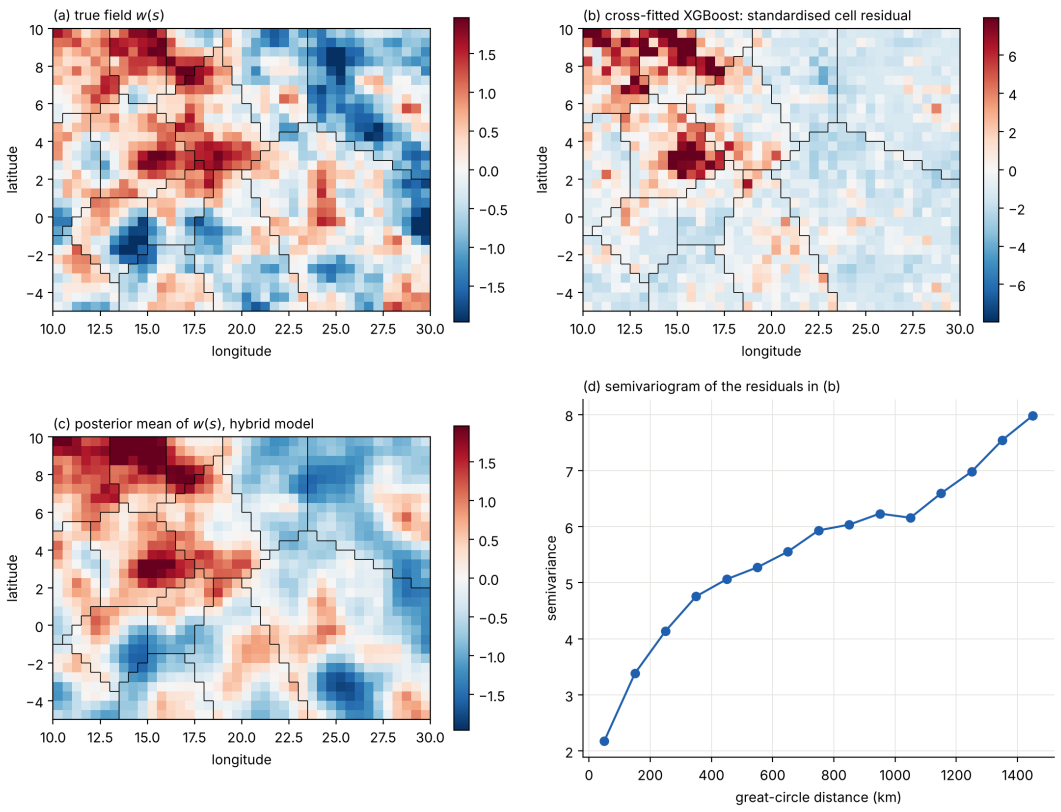


FIG 2. *S1, one replicate. From left: the true field w; standardised cell residuals of the XGBoost model cross-fitted by country (training period); posterior mean of w in the hybrid model (3.6); semivariogram of the XGBoost cell residuals. Colour scales are symmetric and capped at the 98th percentile of the absolute values. Lines are country borders.*

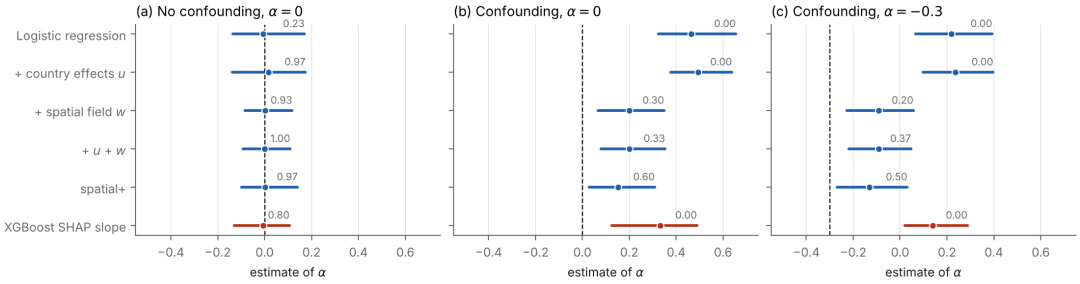


FIG 3. *S2: estimates of the effect α of a national indicator over 30 replicates per scenario. Dots are means and lines span the 5th to 95th percentiles of the estimates across replicates; the number above each line is the coverage of the nominal 95% intervals. Blue: latent Gaussian models; red: the XGBoost summary of the original analysis (least-squares slope of the SHAP values, block-bootstrap interval). The dashed line marks the true value. Full results are in Table S4 of the Supplementary Material.*

regression, but its 95% interval still excludes zero in 20% of replicates (6 of 30; Monte Carlo SE 0.07). Its bootstrap standard error is about 0.70 times the empirical SD (0.044 against 0.063). Its interquartile range, the criterion used in the original analysis, excludes zero in 50% of replicates although Z_0 has no effect. This is the rate expected by construction: the

interquartile range is a 50% interval, so as a test it has a nominal false-positive rate of 50%. It is not a significance criterion.

Under confounding with $\alpha = 0$, no estimator is unbiased:

- logistic regression and the country-effects model report $\hat{\alpha}$ of 0.47 and 0.49 and always reject;
- the spatial field removes about 60% of the bias ($\hat{\alpha} = 0.20$), and spatial+ removes about 70% (0.15, coverage 0.60);
- the spatially adjusted estimators still reject $\alpha = 0$ in 70% (field), 67% (field and country effects) and 40% (spatial+) of replicates (Monte Carlo SE about 0.09);
- XGBoost reports a SHAP slope of 0.33, with an interquartile range that excludes zero in every replicate.

With a true protective effect ($\alpha = -0.3$), logistic regression, the country-effects model and XGBoost report effects of the *wrong sign* ($+0.22$, $+0.24$ and $+0.14$). The XGBoost interquartile range excludes zero in 97% of replicates. The spatial estimators recover the sign on average but only 30–45% of the magnitude (-0.09 to -0.13).

The lesson for the national sustainability indicators is that spatial adjustment reduces confounding bias but does not remove it. Under a confounded null, the adjusted estimators still reject in 40–70% of replicates, so an association that survives spatial adjustment is not evidence of an effect, and the data do not identify its causal size. A large SHAP contribution of a national indicator, stable across block-bootstrap replicates, is fully compatible with no effect at all. These results use a single confounder that varies at the country scale.

The block bootstrap resamples spatial blocks, but Z_0 varies only between countries, and a country spans several blocks. The bootstrap therefore does not reproduce the sampling variability of a country-level effect, whose effective sample size is the number of countries (Section 3.7). This is consistent with the bootstrap standard error being about 30% too small in the scenario without confounding (with 30 replicates, the empirical SD has a relative Monte Carlo error of about 13%).

4.5. *S3: state dependence versus persistent heterogeneity.* Three data-generating processes on a smaller 16×14 domain (224 cells, 6 countries, 104 training weeks, 20 replicates each) produce autocorrelation in Y_{st} for different reasons:

- *state dependence*: $\delta = 1$ and a weak field ($\sigma_w = 0.3$);
- *heterogeneity*: no state dependence ($\delta = 0$), a strong static field ($\sigma_w = 1.3$) and a space-time AR(1) field ($\sigma_v = 0.7$, $\phi = 0.9$ per week, $\rho_v = 300$ km);
- *mixed*: $\delta = 0.5$, $\sigma_w = 1$ and a space-time field with $\sigma_v = 0.5$ and $\phi = 0.9$.

Two further truths repeat the heterogeneity and mixed scenarios with a more persistent space-time field ($\phi = 0.97$ per week). We fit model (3.1) with lags only, with lags and w, and with lags, w and v. The space-time field v is resolved at weekly steps ($\Delta = 1$, the resolution of the data-generating process), at four-week steps ($\Delta = 4$) and at eight-week steps ($\Delta = 8$, the resolution of the regional state-dependence fits in the application). Table 4 shows the three main truths at $\Delta = 1$ and $\Delta = 4$. The $\Delta = 8$ fits, the two persistent truths and the fit times are reported in the text only. We also report the share of total mean absolute SHAP value that the unweighted XGBoost (300 trees, not the original procedure) assigns to the lag feature.

Under state dependence, ignoring the latent field inflates $\hat{\delta}$ by about a quarter (1.26; coverage 0.60), and all latent specifications are approximately unbiased (0.98–1.03) with coverage of 0.90–0.95. Under heterogeneity the contrast is much sharper. With lags only, $\hat{\delta} = 1.61$ although the true value is zero. Adding the static field reduces the estimate to 0.46, but no interval covers zero: slowly varying episodes act as state dependence. Only the models with

the space-time field bring the estimate close to zero: $\hat{\delta}$ is 0.05 with four-week steps and 0.01 with weekly steps, both within Monte Carlo error of zero (Monte Carlo SE about 0.035). The mixed scenario behaves in the same way. The static field leaves an upward bias (0.78 against 0.5), which the space-time field removes (0.55 and 0.54, Monte Carlo SE about 0.04).

The standard errors of $\hat{\delta}$ are too small when v is fitted. In the heterogeneity and mixed scenarios the empirical SD exceeds the mean standard error by 27–43% (ratios 1.27–1.43), and coverage is 0.80–0.90 (Monte Carlo SE 0.07–0.09). In the state-dependence scenario, where the truth has no v, the ratio is 0.92–1.02 for the four latent models. With 20 replicates each ratio has a Monte Carlo SE of about 0.2, so the size of the shortfall is imprecise, but it appears in all twelve cells with v in the truth and in the fitted model (ratios 1.19–1.66, including the $\phi = 0.97$ truths and $\Delta = 8$, with coverage 0.80–0.90). Plug-in intervals for δ from the space-time model are therefore too short, by a factor of roughly 1.2 to 1.7 in these designs. The ratio is not stable enough to justify a single correction factor. In the application, plug-in standard errors from models with a space-time field should be read as too small, and an estimate about two plug-in standard errors from zero should not be read as clearly different from zero.

Resolving v at four-week steps instead of weekly steps changes $\hat{\delta}$ little in these scenarios, and it is four to six times faster (ratios of mean fit time 3.9 to 5.6). Paired by replicate, the four-week estimate is larger by 0.034 (SE 0.006) under heterogeneity and by 0.010 (SE 0.008) in the mixed scenario. The space-time field has an AR(1) coefficient of 0.9 per week, so it changes slowly relative to the four-week history window. This argument is weaker at eight-week steps, since $0.9^8 \approx 0.43$. With $\Delta = 8$ the estimate is still unbiased under state dependence (0.98, coverage 0.95), but under heterogeneity it is 0.10 (empirical SD 0.14, mean SE 0.12, coverage 0.80), about three Monte Carlo SE above zero. Paired by replicate, it exceeds the weekly estimate by 0.086 (SE 0.012). In the mixed scenario it is 0.57 against 0.5 (coverage 0.85; 0.031 above the weekly estimate, SE 0.011). Coarse steps therefore leave part of the episodes to be absorbed by the lag. The bias is smaller when the episodes are more persistent. With $\phi = 0.97$, $\hat{\delta}$ at $\Delta = 8$ is 0.04 under heterogeneity (coverage 0.90) and 0.50 in the mixed scenario (coverage 0.85), and it exceeds the weekly estimate by 0.050 (SE 0.007) and 0.024 (SE 0.006). In these slow truths the static field alone leaves less bias than before (0.38 and 0.71, against 0.46 and 0.78), and the four-week and weekly estimates are within 0.03 of the truth. A fit at $\Delta = 8$ takes 23–27 seconds, against 84–101 at $\Delta = 4$ and 350–474 at $\Delta = 1$. Faster space-time dynamics ($\phi < 0.9$) are not covered, and neither are the 13-week steps of the joint and space-time fits in the application.

Finally, the XGBoost importance of the lag feature is about the same under all truths (share 0.09–0.12, Monte Carlo SE about 0.01), including $\delta = 1$ and $\delta = 0$. With nine features, this is close to an equal share. This configuration of boosting (300 trees, minimum child weight 30, about 23,000 rows at a prevalence of 2.5–4%) does not respond to state dependence at all, and it does not reproduce the dominance of recent history seen in the data. This comparison is therefore uninformative, and we draw no conclusion from it. The point that importance alone cannot show that violence begets violence is a logical one: S3 shows that the lag coefficient itself is strongly inflated by persistent heterogeneity, and an importance measure has no mechanism that would separate the two.

4.6. *S4: biome-specific climate effects.* The temperature effect follows (3.5), with biome means $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$ and a within-biome Matérn deviation η ($\sigma_\eta = 0.15$, $\rho_\eta = 600$ km), $\sigma_w = 0.8$ and 30 replicates. We compare:

- fixed biome-specific slopes, without and with $u + w$;
- the SVC model (3.5);

TABLE 4

S3: estimation of the state-dependence parameter δ (20 replicates per scenario), and the share of the XGBoost mean absolute SHAP value assigned to the lag feature.

Estimator	Mean $\hat{\delta}$	Emp. SD	Mean SE	Coverage
State dependence ($\delta = 1$)				
Lags only	1.263	0.107	0.136	0.60
Lags + w	1.032	0.131	0.139	0.90
Lags + w + v (4-week steps)	0.990	0.135	0.140	0.95
Lags + w + v (weekly steps)	0.997	0.143	0.140	0.95
XGBoost: share of mean |SHAP| for lags	0.108	0.034	–	–
Heterogeneity ($\delta = 0$)				
Lags only	1.614	0.386	0.099	0.00
Lags + w	0.464	0.093	0.111	0.00
Lags + w + v (4-week steps)	0.045	0.152	0.119	0.90
Lags + w + v (weekly steps)	0.011	0.162	0.120	0.90
XGBoost: share of mean |SHAP| for lags	0.094	0.038	–	–
Mixed ($\delta = 0.5$)				
Lags only	1.659	0.351	0.106	0.00
Lags + w	0.782	0.111	0.115	0.25
Lags + w + v (4-week steps)	0.545	0.152	0.120	0.85
Lags + w + v (weekly steps)	0.535	0.172	0.120	0.80
XGBoost: share of mean |SHAP| for lags	0.116	0.044	–	–

- two XGBoost summaries fitted to the anomaly: the per-biome least-squares slope of the SHAP value of the anomaly on the anomaly, and the same slope computed from the SHAP value plus the interaction values credited to the anomaly ($\Phi_{jj} + 2\sum_{k \neq j} \Phi_{jk}$ in the notation of Lundberg et al., 2020). For a pure biome-by-temperature interaction $\gamma_b z$ with biome independent of z, the interventional SHAP value of z is $(\bar{\gamma} + \gamma_b)(z - \mathbb{E}z)/2$, where $\bar{\gamma}$ is the cell-weighted mean of γ_b. The first summary therefore targets $(\bar{\gamma} + \gamma_b)/2$, which equals $\gamma_b/2$ here only because the biomes have equal sizes and $\boldsymbol{\gamma}$ is symmetric about zero. The second targets γ_b in general. A further run with $\boldsymbol{\gamma} = (0, 0.1, 0.2, 0.4)$ checks this;
- the statistic of the original analysis: the median SHAP value of raw temperature within each biome, from a model fitted to raw temperature.

The XGBoost models are original-like. As in the original procedure, each estimate is the median over 20 replicates of the 2.5° block bootstrap on a fixed 30% subsample, and the 95% intervals are bootstrap percentiles. With 20 replicates these are close to the range of the replicates. For the median SHAP value we report the interquartile range of the bootstrap replicates, the criterion of the original analysis. We also report the same statistics from a single fit to all training rows. Coverage is reported for both the biome mean γ_b and the realised biome average $\bar{\beta}_b$ (Section 3.6).

In the symmetric design, the SVC model is the only estimator with close to nominal coverage for the biome means γ_b (0.83–0.97, Monte Carlo SE 0.03–0.07; Figure 4 and Table S5 of the Supplementary Material), although the value of 0.83 in biome 4 is about 1.7 Monte Carlo SE below 0.95. In the asymmetric design its coverage is 0.87, 0.83, 0.77 and 0.77 (Monte Carlo SE 0.06–0.08). This is still the highest coverage in most biomes, but it is below nominal, and in biome 1 the interaction-credited SHAP slope covers in a similar share of replicates (0.80). Its intervals are conservative for the realised biome averages $\bar{\beta}_b$ (coverage 1.00), as expected, because they also account for the sampling of regions within biomes. Fixed biome slopes are nearly unbiased, but their intervals ignore heterogeneity within biomes. They cover γ_b in only 50–60% of replicates (Monte Carlo SE about 0.09), and $\bar{\beta}_b$ in 53–90%. The price

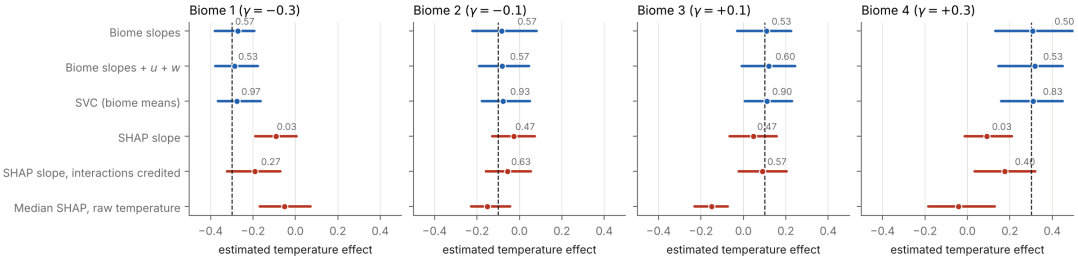


FIG 4. *S4: estimates of the biome-specific temperature effects over 30 replicates. Dots are means and lines span the 5th to 95th percentiles of the estimates across replicates; the number above each line is the coverage of the nominal 95% intervals for γ_b (not shown for the median SHAP value, which has no corresponding estimand). The ordinary SHAP slope targets $(\bar{\gamma} + \gamma_b)/2$, not γ_b, so its low coverage for γ_b in biomes 1 and 4 holds by construction; its coverage of its own target is given in the text. Blue: latent Gaussian models; red: XGBoost summaries. The dashed line marks γ_b. Full results are in Tables S5 and S6 of the Supplementary Material.*

of honest intervals is power: the SVC model detects the small effects ($|\gamma_b| = 0.1$) in only 23–33% of replicates. The SVC model also recovers the coefficient surface (Table S6 of the Supplementary Material). The correlation between $\hat{\beta}(s)$ and $\beta(s)$ is 0.92, the RMSE is 28% lower than with biome slopes only (0.106 against 0.147), and $\hat{\sigma}_\eta = 0.13$ against a true 0.15.

The XGBoost summaries are less informative. Each must be judged against its own target:

- *SHAP slopes.* The per-biome slopes of the ordinary SHAP values have the right ordering but are between 0.26 and 0.48 times γ_b (about 0.3 in three of the four biomes), below the $\gamma_b/2$ expected in this design from the splitting of interactions. Crediting the interaction values to temperature raises them to between 0.56 and 0.89 times γ_b. The remaining attenuation may come from this configuration of boosting (a 30% subsample of about 37,000 rows, a minimum child weight of 30 and early stopping) rather than from boosting in general. With a single fit to all training rows, the slopes are between 0.19 and 0.53 times γ_b for the ordinary SHAP values and between 0.53 and 0.88 times γ_b with the interactions credited. This fit keeps the minimum child weight of 30, which was chosen for millions of rows, so it does not separate the two causes, and we have not done so. Against its own target $(\bar{\gamma} + \gamma_b)/2$, the ordinary SHAP slope covers in 53–63% of replicates (60–67% in the asymmetric design). Fixed biome slopes with the field cover γ_b in a similar share (53–60%; 47–67% asymmetric), so the two are about equally poorly calibrated for their own targets. The interaction-credited slope, which targets γ_b, covers it in 27%, 63%, 57% and 40% of replicates (80%, 67%, 63% and 43% asymmetric), so it under-covers most in the outer biomes. With the asymmetric $\boldsymbol{\gamma} = (0, 0.1, 0.2, 0.4)$ the targets $(\bar{\gamma} + \gamma_b)/2$ are 0.09, 0.14, 0.19 and 0.29. The ordinary SHAP slopes are 0.11, 0.15, 0.19 and 0.23 (Monte Carlo SE about 0.012), so they follow this target in the first three biomes and fall short in the fourth. In biome 1, where temperature has no effect, the slope is positive, as its target is, and its interval excludes zero in 70% of replicates. A reader who takes it as the effect of temperature in that biome would report an effect that does not exist. With the interactions credited the slopes are 0.05, 0.12, 0.20 and 0.29, close to γ_b except in biome 4 (0.71 times γ_b).
- *Median SHAP values.* The median SHAP value of raw temperature within a biome, the statistic of the original analysis, is negative in *every* biome as a bootstrap median (-0.05, -0.15, -0.15 and -0.04), whatever the true effect. From a single fit to all training rows it is -0.02, -0.12, -0.12 and 0.00. Its bootstrap interquartile range excludes zero with a negative sign in every replicate in biome 3 and in 53% of replicates in biome 4, where the true effects are positive. With the asymmetric $\boldsymbol{\gamma}$ it is negative in the first three biomes

(-0.23, -0.17 and -0.08, for true effects of 0, 0.1 and 0.2), and its interquartile range excludes zero with a negative sign in 97%, 100% and 77% of replicates. Applying the original criterion would therefore report a "negative contribution of temperature" where the effect is positive.

The sign of the median SHAP value does not follow the climatology either, since it is not monotone across biomes. It appears to depend on how the trees split credit between raw temperature and the collinear latitude and biome features, whose levels have no effect in the truth. In either case it does not track the effect of temperature (Section 3.6).

4.7. *S5: escalation versus shared episodes.* In the data, recent unrest in a cell predicts conflict there (Section 5.3). This can mean that unrest escalates into violence. It can also mean that both outcomes respond to the same latent episodes, such as a local political crisis, so that recent unrest is only a proxy for the current state of the episode. S5 asks whether the models of the previous sections can tell the two apart, and whether the joint model of Section 3.5 can.

Design.. The domain is a 16×14 block of 0.5° cells (224 cells), with 12 burn-in weeks, 104 training weeks and 26 test weeks. Conflict C_{st} and unrest U_{st} are generated jointly from

$$\operatorname{logit} P(C_{st} = 1) = b_c + d_c L^c_{st} + \kappa L^u_{st} + a(s) + w_c(s) + e(s,t),$$
$$\operatorname{logit} P(U_{st} = 1) = b_u + d_u L^u_{st} + 0.8\, a(s) + w_u(s) + \lambda\, e(s,t),$$

where L^c_{st} and L^u_{st} are $\log(1+\text{events in the previous four weeks})$ of each outcome, $d_c = d_u = 0.3$, $b_c = -3.9$ and $b_u = -3.6$. The static field a is shared by both outcomes (SD 0.6, range 400 km). The fields w_c (SD 0.5, range 300 km) and w_u (SD 0.5, range 150 km) are specific to each outcome. The shared episodes e follow an AR(1) process in time with coefficient 0.95 per week, whose innovations are Matérn fields with range 300 km, scaled to a marginal SD of 0.8. They enter unrest with loading $\lambda = 0.8$. As in the other simulations, every field is drawn from the continuous Matérn covariance, not from the lattice GMRF. The escalation effect is $\kappa = 0$ (no escalation) or $\kappa = 0.4$ (escalation). Shared episodes raise both outcomes together, so recent unrest predicts conflict even when $\kappa = 0$. The data-generating process has a single shared space-time field and no space-time field specific to either outcome, so the joint model below is correctly specified in this respect. Event weeks make up about 3.4% (no escalation) and 3.7% (escalation) of conflict rows and 3.9% of unrest rows.

We compare four estimators of κ, fitted to the training weeks, each with 20 replicates per scenario:

- logistic regression of conflict on L^c and L^u;
- the same with a static Matérn field;
- the same with a static field and a space-time AR(1) field at four-week steps (Section 3.3);
- the joint model of both outcomes (Section 3.5), with outcome-specific intercepts, lags and static fields, and a space-time field at four-week steps shared by the two outcomes, which enters unrest with an estimated loading λ.

The joint model is not the data-generating process. It has no shared static field, so a must be absorbed by the two outcome-specific fields, and its space-time field is piecewise constant over four weeks while e changes weekly. The application differs from this design: it uses steps of 13 weeks on a 1° lattice, cross-lags in both directions and neighbour histories of both outcomes (Section 5.3). We also forecast conflict over the 26 test weeks and score the forecasts on all test rows and on onset rows, those of cells without a conflict event in the previous 52 weeks.

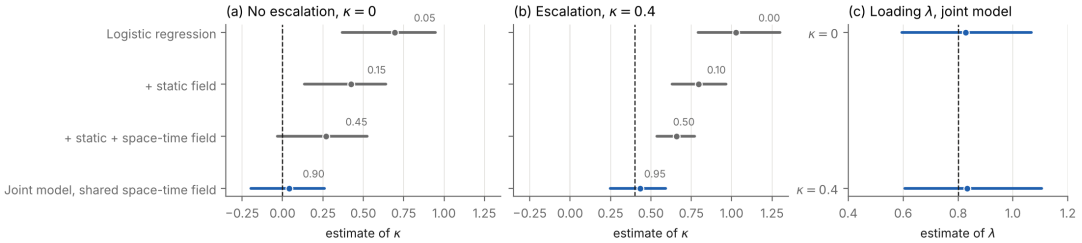


FIG 5. *S5: estimates of the escalation effect κ of recent unrest on conflict (a, b) and of the loading λ of the shared space-time field in the joint model (c), over 20 replicates per scenario. Dots are means and lines span the 5th to 95th percentiles of the estimates across replicates; the number above each line is the coverage of the nominal 95% intervals for κ. Grey: univariate models of conflict; blue: the joint model. The dashed line marks the true value. Full results are in Table S11 of the Supplementary Material.*

Results.. Figure 5 summarises the estimates, and Table S11 of the Supplementary Material gives the full results. Monte Carlo SEs of coverage and rejection rates are at most 0.11 with 20 replicates.

- *No escalation.* All three univariate models report escalation that does not exist. Logistic regression gives $\hat{\kappa} = 0.70$ (empirical SD 0.19), the static field 0.43 (SD 0.16) and the space-time field 0.27 (SD 0.16). They reject $\kappa = 0$ in 95%, 85% and 55% of replicates. The joint model gives 0.04 (SD 0.17, Monte Carlo SE of the mean 0.04), with coverage 0.90 (Monte Carlo SE 0.07) and a false rejection rate of 0.10.
- *Escalation.* The univariate models overestimate $\kappa = 0.4$ (1.03, 0.79 and 0.66), with coverage 0.00, 0.10 and 0.50. The joint model gives 0.43 (SD 0.11, Monte Carlo SE of the mean 0.025), with coverage 0.95 (Monte Carlo SE 0.05), and it rejects $\kappa = 0$ in 95% of replicates.

A space-time field for conflict alone removes most of the bias in the own-lag coefficient d_c (0.37–0.38 against 0.3, compared with 1.52–1.54 for logistic regression and 0.80–0.82 with the static field), but not the bias in κ. A plausible explanation is that conflict events are rare, so a field estimated from conflict alone tracks the episodes poorly, and recent unrest still carries information about the current episode. The joint model estimates the field from both outcomes. Its estimate of d_c is 0.36–0.38.

The joint model also recovers the shared field. The estimated loading has mean 0.83 in both scenarios (SD 0.14 without and 0.15 with escalation) against a true 0.8, and the range of the shared field has mean 315 km (SD 40–44) against 300 km. As in S3, the standard error of the lag coefficient is too small when a space-time field is fitted under no escalation: the empirical SD of $\hat{\kappa}$ is 1.20 times the mean SE (0.17 against 0.14). Under escalation the ratio is 0.89. With 20 replicates these ratios are imprecise, but together with S3 they indicate that plug-in intervals for κ near zero are too short.

The joint model does not forecast conflict better than the univariate latent models. Paired by replicate, its log score differs from that of the univariate space-time model by $+0.0002$ (SE 0.0005) without escalation and $+0.0003$ (SE 0.0005) with escalation, and its AUC-PR by -0.002 (SE 0.002) and -0.001 (SE 0.002). Onset AUC-PR is also similar (0.033 against 0.034 without escalation, 0.042 against 0.045 with escalation). The latent models forecast better than logistic regression. Without escalation, for example, the joint model improves on it by -0.003 in log score (SE 0.001) and by $+0.025$ in AUC-PR (SE 0.004). Predictive accuracy therefore cannot show whether unrest escalates into conflict: the models predict equally well and disagree about κ. The cost of the joint model is computational. A fit takes about 310–320 seconds, against about 80 seconds for the univariate space-time model.

These results hold for a single shared episode process whose loading is constant in space, with one direction of escalation. Faster episodes, a loading that varies by region, escalation that runs in both directions, and the 13-week steps of the application are not covered. S5 also cannot detect the opposite failure. If unrest has episodes of its own that do not involve conflict, a joint model with a single shared field must place them in v, which enters the conflict equation with loading one. Conditioning on v then removes part of the variation in the unrest history through which escalation would act, and $\hat{\kappa}$ is biased towards zero. Because both scenarios of S5 contain only shared episodes, this attenuation is not tested here. The recovery of κ in S5 therefore supports the joint model only when unrest has no specific space-time variation. [TODO: biv2 and S5 extension results: S5 scenarios with unrest-only episodes, with and without escalation, fitted with the single-field joint model and with the extended model that has an additional unrest-specific space-time field.]

5. Application to ACLED 2018–2024. All models are trained on 2018–2022 and evaluated on 2023–2024, except where we state that they are updated during the test period. The boosting models use the hyperparameters of Menicali, Marcantonio and Castruccio (2026). The test period includes the onset of new large wars, in Sudan from April 2023 and in Gaza from October 2023, which a model trained on 2018–2022 cannot anticipate from recent history. Besides the scores on all test rows, we therefore report scores on onset rows: test cell-weeks in cells with no event of that outcome in the previous 52 weeks (Section 3.10). The computations were run at the Notre Dame Center for Research Computing.

5.1. *Reconstructing the cell-week table.* The table built by the original pipeline has 70,033,444 rows but only 34,671,180 distinct cell-weeks (Section 2.1). In one partition, 613,459 cell-weeks appear once, while others appear 2, 3, 8, 16, 32, 48 and up to 648 times. The national indicators and IPC regions are attached by spatial intersection and then merged on cell and year. A cell that touches several countries or IPC regions is therefore multiplied at every join, and each copy combines indicators from different units.

The duplication matters for more than the covariates. The four-week event counts of the original pipeline are computed row by row within each cell, after sorting by cell and week. When a cell-week has copies, the rows immediately before a given row are copies of the same week, so the "past" count contains the current week's events, which are the outcome. We recomputed the original counts exactly and compared them with the true counts over the four previous weeks, on 69.4 million rows of the duplicated table, over all years and without the coverage filter. Among rows with an event, counting each copy separately, 22% have an original count that includes the current week's events while the true past count is zero, for both outcomes. Used directly as a predictor of the outcome on these rows, without any model, the original count reaches an AUC-PR of 0.883 for conflict and 0.833 for unrest. The true count reaches 0.495 and 0.450. These values describe the duplicated table as a whole. They are not on the same rows as the test-period scores below. The leak arises only in cell-weeks that have copies, and never in the first copy of a cell-week, which is the one the repaired table keeps. Copies come from cells that intersect several countries or several IPC regions; we have not checked how many of them lie on international borders. [TODO: Share of duplicated cells that intersect an international border, against those that cross only IPC boundaries.]

The repaired table has one row per cell-week (Section 2.1). After ACLED coverage and the biome exclusion it has 19,364,639 cell-weeks in 59,339 cells and 366 weeks. The test period has 4,787,224 cell-weeks in 46,031 cells with complete climate records.

5.2. *The boosting analysis before and after the repair.* Table 5 reports three versions of the XGBoost analysis. The first, *as published*, applies the original procedure to the duplicated table: the climate variables of the current week and their four-week means as features, positives weighted by $(n_0/n_1)^{1/2}$, 5 million training rows sampled with replacement, and the number of trees chosen by early stopping on the test years. The second, *deduplicated*, applies the same procedure unchanged to the repaired table. It isolates the effect of the duplication and the leak. The third, *corrected*, also changes the procedure. The number of trees is chosen by early stopping on 2022 after training on 2018–2021, and the model is then refitted on 2018–2022. It uses no class weight. Its features use only information available before the week, so the climate variables of the current week are dropped and only their four-week means are kept. The 5 million training rows are sampled without replacement. The weighted models are prior-corrected before scoring (Section 3.8). We also fit the location-aware model of Section 3.10, which adds each cell's training event rate to the corrected model. For the fit that chooses the number of trees, this rate uses 2018–2021 only; for the final fit it leaves out the row's own year.

On the duplicated table we recover the published performance, with AUC-PR 0.934 for conflict and 0.917 for unrest, against 0.93 and 0.91 reported. Split by cell, nearly all of it comes from the cells that have copies in the original table (AUC-PR 0.966 and 0.968), and in the other cells AUC-PR is 0.498 and 0.528. The first two rows of Table 5 are on different test sets: the duplicated table has 14,425,130 test rows with conflict prevalence 2.98%, the repaired table 4,787,224 rows with 1.62%. AUC-PR depends on prevalence, so part of the difference between them is not due to the model. Scored on one row per cell-week of the repaired test set, the model fitted as published reaches AUC-PR 0.473 for conflict and 0.479 for unrest. The deduplicated and corrected rows are on identical test rows and can be compared directly. Deduplication alone gives AUC-PR 0.546 for conflict and 0.555 for unrest. The corrected model reaches 0.544 and 0.557, the persistence baseline 0.481 and 0.468, and the location-aware model 0.566 and 0.583. The scores are similar in 2023 and 2024. On onset rows, where the prevalence is 0.10% for conflict and 0.14% for unrest, the corrected model reaches AUC-PR 0.0073 and 0.0097, and the location-aware model 0.0111 and 0.0119.

Three conclusions follow. First, a flexible model of history and covariates improves on persistence, but by less than the published scores suggest: the corrected model improves on persistence by 0.06 in AUC-PR for conflict and 0.09 for unrest. This gain cannot be attributed to the covariates alone. The persistence baseline uses only the total count of the outcome, while the corrected model also uses the five event types separately, including those of the other outcome, in a nonlinear way, together with calendar indicators and country-year indicators whose joint values can identify a country. [TODO: XGBoost on the event histories alone, with the same tuning.] Second, a cell intercept adds a further 0.02 to 0.03, so part of what the covariates leave unexplained is persistent and local. Third, on onset rows all models rank cell-weeks several times better than chance, but their AUC-PR stays near 0.01. New events in places without recent events are rarely anticipated.

5.3. *Escalation versus shared episodes.* Recent unrest predicts conflict. In every univariate model of conflict fitted here, the coefficient of the unrest history is positive and more than two standard errors from zero (Section 5.6 and Table 6). S5 (Section 4.7) showed that this is weak evidence of escalation. When the two outcomes share latent episodes, univariate models report escalation that does not exist, even when they include their own space-time field. The joint model of Section 3.5 recovered the escalation coefficient κ in both scenarios of S5, and we fit it to the data. The application lies outside the S5 design in several respects. S5 as reported has a single shared space-time field and no outcome-specific space-time field, fits that field at steps of $\Delta = 4$ weeks on the 0.5° lattice, has escalation in one direction only

TABLE 5

XGBoost on the 2023–2024 test period. As published: duplicated table, original procedure (current-week features, class weight, early stopping on the test years). Deduplicated: repaired table, original procedure. Corrected: repaired table, features available before the week, no class weight, early stopping on 2022. Corrected + cell event rate: the location-aware model. AUC-PR is also reported separately for cells that have copies in the original table and for the others. The persistence baseline ranks cell-weeks by the number of events in the previous four weeks. The as-published rows are on a different test set from the other rows, with a different prevalence.

Pipeline	Outcome	Test rows	Prevalence	Trees	AUC-PR	AUC-PR by cell	
						dup. in original	other cells
As published	Conflict	14.43M	2.98%	1998	0.934	0.966	0.498
As published	Unrest	14.43M	3.58%	1474	0.917	0.968	0.528
Original procedure, repaired table	Conflict	4.79M	1.62%	488	0.546	0.590	0.523
Original procedure, repaired table	Unrest	4.79M	2.73%	646	0.555	0.543	0.558
Corrected	Conflict	4.79M	1.62%	637	0.544	0.588	0.521
Corrected	Unrest	4.79M	2.73%	814	0.557	0.545	0.559
Corrected + cell event rate	Conflict	4.79M	1.62%	381	0.566	0.598	0.549
Corrected + cell event rate	Unrest	4.79M	2.73%	135	0.583	0.567	0.586

and has no neighbour histories. The application uses steps of 13 weeks on a 1° lattice, cross-lags in both directions and the neighbour histories of both outcomes. A single shared field also carries a risk that S5 cannot detect. Episodes of unrest that do not involve conflict are forced into v, which enters the conflict equation with loading one. Conditioning on v then removes part of the variation in the unrest history through which escalation would act, and $\hat{\kappa}$ is biased towards zero. [TODO: biv2 and S5 extension results: joint model with an additional unrest-specific space-time field, and S5 scenarios with unrest-only episodes, with and without escalation.]

Region and models.. The region is West Africa and the Sahel, longitude 18°W to 16°E and latitude 4°N to 25°N. It has 2,505 cells in 22 countries and 906,810 cell-weeks per outcome with complete history. It contains the central Sahel, the Lake Chad basin and the coastal states of West Africa. The shared field v is defined on 641 cells of 1°, with 28 steps of $\Delta = 13$ weeks. We fit three models, with the covariates, initial-conditions term, year effects and missingness indicators of Section 5.6, and the own and neighbour histories of both outcomes:

- *univariate static*: conflict only, with country effects, cell effects and a static Matérn field;
- *univariate space-time*: the same, plus a space-time field for conflict alone;
- *joint*: model (3.4), in which the space-time field is shared by the two outcomes.

The second model is the joint model with $\lambda = 0$. Comparing the two shows the effect of letting unrest inform the space-time field, under the assumption that there is no unrest-specific space-time variation. Table 6 reports the lag coefficients, the hyperparameters of the space-time fields and the test scores. Table S12 of the Supplementary Material gives all hyperparameters and coefficients.

Results.. In the univariate static model the coefficient of recent unrest on conflict is $\hat{\kappa} = 0.138$ (SE 0.031). Adding a space-time field for conflict hardly changes it (0.131, SE 0.032), although the field is clearly present (range 423 km, standard deviation 0.69, AR(1) coefficient 0.90 per 13 weeks) and it reduces the own-lag coefficient from 0.575 (SE 0.016) to 0.444 (SE 0.017). This is the pattern of S5: a field estimated from conflict alone absorbs part of the persistence of conflict, but not the information about the current episode that is carried

TABLE 6

Conflict and unrest in West Africa and the Sahel: lag coefficients (standard errors) of the conflict equation, hyperparameters of the space-time field, and test scores for 2023–2024. Own: conflict in the previous four weeks; other: unrest in the previous four weeks (κ); neighbour terms: the same histories averaged over the rook neighbours of the cell. In the joint model the space-time field is shared by the two outcomes and enters unrest with loading λ; its AR(1) coefficient is per step of 13 weeks. Full results in Table S12 of the Supplementary Material.

Region and model	$\hat{\kappa}$ unrest history	$\hat{\kappa}^{\mathrm{nb}}$ neighbours' unrest	$\hat{\delta}_{\mathrm{C}}^{\mathrm{nb}}$ neighbours' conflict	$\hat{\lambda}$	$\hat{\rho}_v$ (km)	$\hat{\phi}$
West Africa and Sahel						
Conflict only, $u + e + w$	0.138 (0.031)	0.005 (0.008)	0.129 (0.007)	–	–	–
Conflict only, $u + e + w + v$	0.131 (0.032)	0.003 (0.008)	0.051 (0.008)	–	423	0.90
Joint, shared v	0.044 (0.032)	−0.020 (0.008)	0.061 (0.008)	0.87	499	0.95
Joint, shared v + unrest-specific v_{U}	[TODO: biv2 results]					
AME region: [TODO: running]						

by recent unrest. In the joint model the coefficient falls to 0.044 (SE 0.032). The plug-in interval of ± 1.96 SE runs from -0.018 to 0.106, but it is likely too short. In S5 without escalation, the empirical SD of $\hat{\kappa}$ from the joint model was 1.20 times its mean SE, and in S3 the corresponding ratio for $\hat{\delta}$ was 1.19 to 1.66 whenever a space-time field was fitted (Sections 4.7 and 4.5). The own-lag coefficient stays at the level of the univariate space-time model (0.459, SE 0.017). The coefficient of unrest in neighbouring cells is close to zero in the univariate models (0.005 and 0.003, SE 0.008) and negative in the joint model (-0.020, SE 0.008).

The shared field has loading $\hat{\lambda} = 0.87$ on unrest, range 499 km, standard deviation 1.01 and AR(1) coefficient 0.95 per 13 weeks. It is a shared latent component of recorded risk that varies over several hundred kilometres and changes slowly over quarters; what it represents is discussed below. In the unrest equation of the joint model, the coefficient of recent conflict on unrest is 0.066 (SE 0.031), and that of conflict in the neighbouring cells is -0.042 (SE 0.013). Negative neighbour coefficients in both equations would mean that events in neighbouring cells lower the risk in a cell, which is implausible as a substantive spillover. One possible reading is that the shared field over-absorbs regional variation, and the neighbour terms compensate. We take them as a possible sign of misspecification of the joint model.

The joint model does not forecast conflict better. Its test log score for conflict is 0.1031, against 0.1020 for the univariate static model and 0.1026 for the univariate space-time model, and its AUC-PR is 0.466 against 0.470 and 0.471, at a conflict prevalence of 4.2%. We have not computed standard errors for these differences, so we do not read their order as evidence for one model. As in S5, the three models predict conflict similarly and disagree about κ. Predictive accuracy cannot decide between escalation and shared episodes. The joint fit took about 4.5 hours, against 8 minutes for the univariate static model and 43 minutes for the univariate space-time model.

Interpretation.. Under this model, and pending the extended model described above, the univariate association between recent unrest and conflict in this region is largely accounted for by shared episodes. Under no escalation, the univariate space-time model rejected $\kappa = 0$ in 55% of the S5 replicates, while the joint model had coverage 0.90. A univariate estimate of 0.13 to 0.14 is therefore not, by itself, evidence that unrest escalates into conflict. The joint estimate does not show that escalation is absent. Its interval includes values of about 0.1 even before allowing for the SE shortfall above. The fall from 0.131 to 0.044 also has

a second reading. Compared with the conflict-only field, the shared field has a larger standard deviation (1.01 against 0.69) and range (499 against 423 km). This is what removing confounding by shared episodes would produce, but also what a field that absorbs unrest-specific episodes would produce, with $\hat{\kappa}$ attenuated towards zero. The two readings cannot be separated with a single shared field, and we leave the conclusion open until the extended model is fitted. [TODO: biv2 and S5 extension results.] Three further limits apply. First, κ is the short-run, local part of escalation. Escalation that unfolds over several months and over hundreds of kilometres cannot be told apart from a shared episode, and the joint model attributes it to v (Section 3.5). The finding is thus compatible with accounts in which repression of protest leads to armed conflict over longer periods (Ritter and Conrad, 2016; Ryckman, 2020; Chenoweth, Perkoski and Kang, 2017; Rød and Weidmann, 2023). It indicates that four-week unrest histories in the same cell are a weak signal of such escalation once shared episodes are modelled. Second, this is also why unrest histories help forecasting models of conflict (Rød, Hegre and Leis, 2025): they are informative about the current episode, whatever its nature. Third, the shared episodes mix occurrence and reporting. Reporting of protest clusters in space and time (Hellmeier, Weidmann and Rød, 2018), and the reporting of non-fatal events depends on attention and sources (Croicu and Eck, 2022). A period of media attention to a region raises the recorded counts of both event types, and the joint model treats it as a shared episode. With a single event source the two cannot be separated, so neither v nor λ should be read as a property of occurrence alone.

The AME region.. [TODO: Joint fit for the AME region (13,719 cells, v on 3,637 cells of 1°): $\hat{\kappa}$ and SE in the univariate static, univariate space-time and joint models; $\hat{\lambda}$, range, standard deviation and AR(1) coefficient of the shared field; test scores. The fit is still running.]

5.4. *Residual spatial structure and the hybrid model.* We fit the hybrid model (3.6) to the AME region, that is Africa, the Middle East and adjacent regions (longitude 20°W to 65°E, latitude 36°S to 45°N; 13,719 cells in 93 countries, 4,976,634 cell-weeks of which 1,426,776 are in the test period; Section 2.1). It combines the XGBoost offset cross-fitted by country, country effects, and either a Matérn field with sea barriers (Remark 1) or independent cell effects. The offset is trained on the rows of this region only, whereas the corrected XGBoost model of Section 5.2 is trained on all locations worldwide. The comparisons below are on identical test rows, but the training sets differ.

Table S7 of the Supplementary Material reports the test-period residual diagnostics of every model, and Figure S1 of the Supplementary Material the semivariograms of the test-period residuals. The residuals of the corrected XGBoost model have Moran's I of 0.42 for conflict and 0.15 for unrest (Section 2.4). The location-aware model reduces it to 0.29 for conflict but increases it to 0.25 for unrest. The hybrid models do not reduce the overall test-period residual structure. For conflict, Moran's I is 0.45 with the field or with independent cell effects, about the same as for the corrected model. For unrest it is 0.29, about twice the value of the corrected model and above the 0.26 of the cross-fitted offset alone. For unrest, latent effects estimated on the training years therefore increase the test-period residual autocorrelation. One possible explanation is that cell effects estimated in 2018–2022 are carried into a period in which recorded unrest rose and its cell-level pattern changed (Section 2.1); the location-aware model shows a similar increase. The hybrid models do reduce the residual correlation across borders, from 0.21 to 0.14 for conflict and from 0.08 to -0.02 for unrest. Residual structure in the test period is therefore not removed by latent effects estimated on the training years. Section 5.5 examines why. For the cross-fitted offset, the across-border correlation is 0.25 for pairs of countries in the same fold and 0.28 for pairs in different folds for conflict, and 0.14 and 0.09 for unrest. The same-fold pairs are few (264 of 2,040).

Table S8 of the Supplementary Material gives the fitted hyperparameters. For conflict, the field has range 263 km and standard deviation 1.48. For unrest, the range is 65 km and the standard deviation 1.25. A range close to the size of a cell, about 55 km at the equator, means that the field is not resolved by the grid (Section 3.2). The field then behaves like independent cell effects, and it should not be read as regional hotspots that cross borders. This is the case for unrest. The field improves the log marginal likelihood over independent cell effects by 1,254 for conflict but only by 95 for unrest, and the two give almost the same test scores for unrest (Table 7). For unrest, cell-level effects would largely track towns and cities, where protests are both more frequent and more often reported. A fit to all locations worldwide (46,031 cells) gives a range of 313 km for conflict and 65 km for unrest. The unrest results of the AME region also depend on its composition. It includes Mediterranean Europe, where unrest is dense and where several countries enter ACLED only in 2020 (Section 2.1). [TODO: Sensitivity of the unrest range, the hotspot shares and the national coefficients to restricting the region to Africa and the Middle East proper.]

Figure 6 compares observed and predicted frequencies over the test period. The hybrid model reproduces the regional pattern of both outcomes, including the Sahel, eastern Congo, Somalia, Yemen and Syria–Iraq for conflict. For both outcomes the hybrid model is close to the diagonal above a predicted probability of about 1% and overpredicts below it, where the observed frequency is lower than predicted. The cross-fitted offset alone underpredicts in the same bins.

The estimated fields are shown in Figure 7. Table S9 of the Supplementary Material summarises the posterior of the field averaged over eleven regions. The regions were chosen before the fit: eight known conflict zones and three regions expected to be quiet. A posterior probability close to one that the average over a known conflict zone is positive is therefore expected and is not a finding in itself. The last two rows of the table give a summary that does not depend on chosen regions: the share of cells whose posterior probability that $w(s) > 0$ exceeds 0.95, and the share for which it is below 0.05. For conflict these shares are 18.9% and 3.0%, and for unrest 11.6% and 2.6%. For conflict, the averages over the three quiet regions, the western Congo basin, the interior of the Arabian peninsula and southern Africa, are negative. For unrest, only the average over eastern Congo is clearly positive, and none is clearly negative. Negative values of w mark low *recorded* risk relative to the covariates, recent history and country effects. In regions with restricted media access and few ACLED sources, such as the interior of the Arabian peninsula, they may reflect reporting rather than occurrence (Section 2.1). Positive values in cells with towns or administrative centres may in part reflect events of low spatial precision, which ACLED places at such centroids. Because the offset is cross-fitted by country, errors of the offset that are common to a country are absorbed by u, but errors that vary within a country can be absorbed by w.

5.5. *Forecasting.* Table 7 compares out-of-sample scores on identical test rows of the AME region. The reference is the corrected XGBoost model. Differences in mean log score are given against it, with standard errors clustered by country (standard errors clustered by cell are smaller). A gain of a hybrid model over the cross-fitted offset alone is not evidence for the latent effects (Section 3.8). The relevant comparisons are with the corrected and location-aware models. The table has two parts. In the static part, every model is fitted once on 2018–2022. In the updated part, the models are updated each quarter of the test period with all data before the start of the quarter, as an early-warning system would be run. The location-aware boosting model is updated by recomputing the cell event rate. The hybrid models keep the hyperparameters fitted on 2018–2022 and recompute the latent effects. Table S10 of the Supplementary Material reports all dynamic experiments.

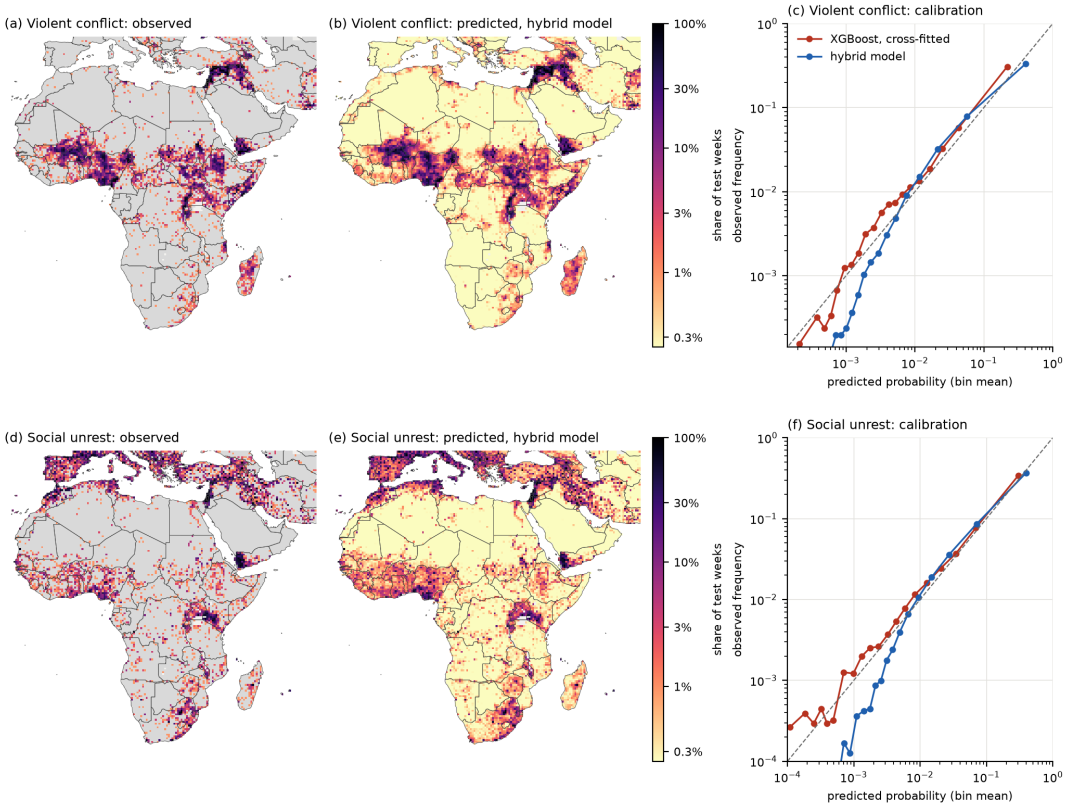


FIG 6. *Observed and predicted event frequency in the test period 2023–2024, AME region. (a, d) Share of test weeks with at least one event in each cell; grey cells have no event week. (b, e) Mean predicted probability of the hybrid model with country effects and the Matérn field over the same weeks, on the same colour scale. (c, f) Calibration: observed frequency against mean predicted probability in 20 equal-count bins, for the hybrid model and the XGBoost offset cross-fitted by country; the dashed line is perfect calibration.*

Static models.. With the field, the hybrid model improves on the corrected XGBoost model in log score by 0.0021 (SE 0.0007) for unrest. For conflict the difference, 0.0012 (SE 0.0009), is within about one standard error. The hybrid models do not reach the location-aware model, which improves on the corrected model by 0.0035 (SE 0.0013) and 0.0032 (SE 0.0008). The Brier score, which Table 7 does not report, points the other way for conflict. The hybrid model with the field has a Brier score higher than the corrected model by 0.0004 (SE 0.0002) for conflict, and lower by 0.0001 (SE 0.0002) for unrest. The location-aware model lowers the Brier score by 0.0003 (SE 0.0002) and 0.0004 (SE 0.0002). The field and the independent cell effects give almost the same scores: for conflict the field is better by 0.0002 in log score, and for unrest by less than 0.0001. A hybrid model with a space-time field ($\Delta = 13$ weeks, forecast through its AR(1) dynamics) improves only slightly on the static field, by 0.0004 for conflict and 0.0003 for unrest. In AUC-PR, the corrected model reaches 0.515 for conflict and 0.532 for unrest, the hybrid model with the field 0.519 and 0.540, the location-aware model 0.533 and 0.550, and persistence 0.446 and 0.451. On onset rows, where the prevalence is 0.15% for conflict and 0.18% for unrest, AUC-PR is 0.008 to 0.011 for conflict and 0.010 to 0.014 for unrest for all models except the cross-fitted offset alone, which reaches 0.006 and 0.009.

Updated models.. Quarterly updating improves every model that is updated; the corrected XGBoost model and the cross-fitted offset are not updated. The updated hybrid model with

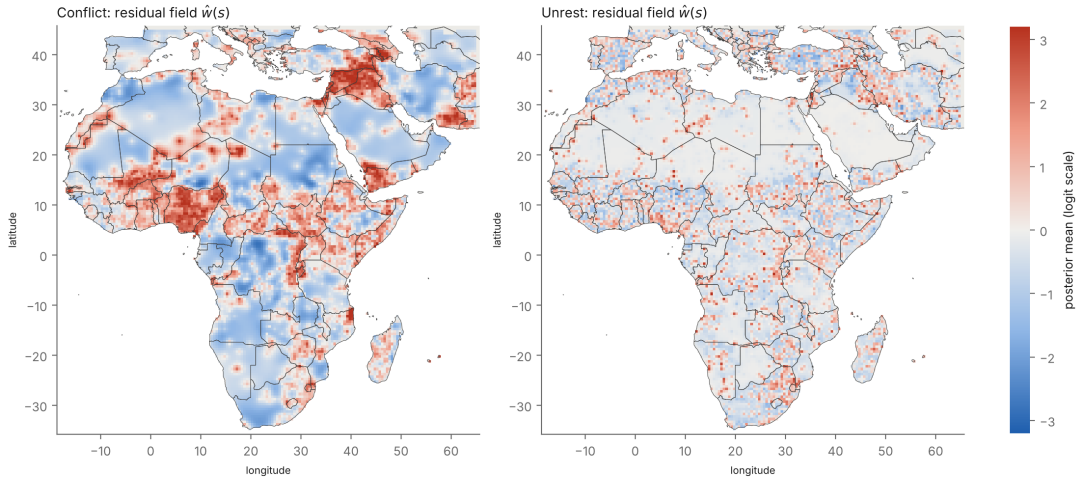


FIG 7. *Posterior mean of the residual field $w(s)$ of the hybrid model for conflict (left) and unrest (right), on the logit scale. Positive values mark higher recorded risk than predicted by the covariates, recent event history and country effects. Recorded risk mixes occurrence and reporting, and events of low spatial precision are placed at town and administrative centroids (Section 2.1).*

the space-time field improves on the corrected XGBoost model by 0.0035 (SE 0.0008) for conflict and 0.0034 (SE 0.0007) for unrest. It thus reaches the score of the static location-aware model, and its AUC-PR is 0.539 and 0.554. It also has the highest AUC-PR on onset rows, 0.0120 for conflict and 0.0142 for unrest. Updating also improves the location-aware model, however, to 0.0050 (SE 0.0013) and 0.0044 (SE 0.0008) in log score, and AUC-PR 0.544 and 0.559. Of the gap in log score between the static hybrid model with the field and the static location-aware model, 0.0023 for conflict and 0.0012 for unrest, a gap of 0.0015 and 0.0010 remains between the best updated models of each kind. In the Brier score the updated space-time hybrid model is better than the corrected model, by 0.0001 (SE 0.0001) for conflict and 0.0004 (SE 0.0002) for unrest, and the updated location-aware model by 0.0005 (SE 0.0002) and 0.0006 (SE 0.0002). The updated space-time hybrid model removes most of the test-period residual structure: Moran's I of its residuals is 0.12 for conflict and 0.10 for unrest, against 0.32 and 0.26 for the updated location-aware model.

We also refitted the hybrid models, including the boosting offset and the hyperparameters, on 2018–2023 and scored 2024. These comparisons are on the 2024 rows only. The refitted hybrid model with the field improves on the corrected model by 0.0022 for conflict and 0.0033 for unrest, against 0.0007 and 0.0017 without the refit. The location-aware model, which is not refitted in this experiment, improves on it by 0.0028 and 0.0023. A year of new data thus brings the hybrid model close to the location-aware model for conflict and ahead of the static location-aware model for unrest.

Where the residual structure lies.. To see why latent effects estimated on the training years help little, we fitted the field and the independent cell effects to the test period itself, with the same offset. This is an in-sample upper bound on what a static latent term can absorb. For conflict, the test-period field has a range of 374 km, the field fits much better than independent cell effects (log marginal likelihood higher by 1,045), and it removes the residual autocorrelation (Moran's I from 0.54 to -0.12, against 0.35 with independent cell effects). Its correlation with the field fitted on 2018–2022 is 0.79. For unrest, the test-period field has a range of 66 km, its advantage over independent cell effects is small (69 in log marginal

TABLE 7

Out-of-sample scores, AME region, 2023–2024, on identical test rows. Static models are fitted once on 2018–2022; updated models are updated each quarter with all data before the quarter (hybrid models: latent effects recomputed with the hyperparameters fixed; location-aware model: cell event rate recomputed). Differences in mean log score against the corrected XGBoost model, with standard errors clustered by country. Lower is better for the log score. The hybrid offset is trained on the AME region only; the corrected and location-aware XGBoost models are trained on all locations. Onset rows: cells without an event of the outcome in the previous 52 weeks. Further experiments in Table S10 of the Supplementary Material.

Model	Conflict			Unrest		
	AUC-PR	Δ log score $\times 10^3$ (SE)	Δ Brier $\times 10^3$ (SE)	AUC-PR	Δ log score $\times 10^3$ (SE)	Δ Brier $\times 10^3$ (SE)
Fitted once on 2018–2022						
XGBoost, corrected (reference)	0.515	reference	reference	0.532	reference	reference
XGBoost + cell event rate	0.533	−3.50 (1.27)	−0.32 (0.22)	0.550	−3.22 (0.78)	−0.36 (0.18)
XGBoost, cross-fitted by country (offset)	0.461	8.24 (2.27)	1.39 (0.42)	0.493	4.74 (1.20)	0.91 (0.28)
Offset + u + iid cell effects	0.519	−0.93 (0.89)	0.42 (0.21)	0.540	−2.04 (0.67)	−0.10 (0.18)
Offset + $u + w$	0.519	−1.17 (0.94)	0.44 (0.23)	0.540	−2.07 (0.66)	−0.10 (0.18)
Offset + u + cell effects + v, 13-week steps	0.519	−1.56 (0.96)	0.30 (0.24)	0.541	−2.33 (0.79)	−0.19 (0.21)
Offset + u + cell effects + v, 26-week steps	0.518	−1.52 (0.95)	0.34 (0.25)	0.540	−2.29 (0.80)	−0.17 (0.22)
Updated each quarter of the test period						
XGBoost + cell event rate	0.544	−4.98 (1.31)	−0.54 (0.23)	0.559	−4.38 (0.83)	−0.59 (0.17)
Offset + u + iid cell effects	0.531	−2.30 (0.76)	0.12 (0.14)	0.550	−3.03 (0.72)	−0.35 (0.18)
Offset + $u + w$	0.531	−2.65 (0.77)	0.11 (0.14)	0.550	−3.06 (0.72)	−0.35 (0.18)
Offset + u + cell effects + v, 13-week steps	0.539	−3.52 (0.81)	−0.14 (0.09)	0.554	−3.39 (0.71)	−0.42 (0.17)
Persistence (past 4 weeks)	0.446	–	–	0.451	–	–

likelihood), and its correlation with the training field is 0.69. Conflict residual risk is therefore smooth at a scale of a few hundred kilometres and partly moves over time, which is why updating, and a space-time field, help. Unrest residual risk is mainly cell-level, which is what a cell event rate captures directly. A location-aware boosting model is a strong benchmark for both.

5.6. *State dependence, climate and national indicators.* The parametric latent models of Section 3.1 are fitted to the AME region (3,494,982 training rows, 13,719 cells). We compare logistic regression without latent effects, a model with country effects, and models with country effects, independent cell effects and a Matérn field, with and without a spatially varying coefficient for maximum temperature and with the spatial+ adjustment of Section 3.7.

State dependence.. The coefficient δ of the own four-week history is 2.04 (SE 0.007) for conflict in logistic regression, 1.75 with country effects and 0.92 (SE 0.008) with country effects, cell effects and the field. For unrest it is 1.62, 1.47 and 0.78. The coefficient of the other outcome's history falls in the same way, from 0.40 to 0.15 for unrest on conflict and from 0.22 to 0.13 for conflict on unrest. In the central Sahel (9°N to 20°N, 6°W to 16°E, a box of 22° by 11°; 968 cells, steps of 8 weeks), adding a space-time field lowers $\hat{\delta}$ further: for conflict from 1.58 without latent effects to 0.65 with the static terms and 0.28 (SE 0.021) with the space-time field, whose range is 230 km and AR(1) coefficient 0.93 per 8 weeks. For unrest the corresponding values are 1.27, 0.42 and 0.17 (SE 0.060), with an AR(1) coefficient of 0.49 per 8 weeks, about 0.915 per week. Much of the apparent state dependence is therefore absorbed by latent risk that varies slowly in space and time. Three caveats apply. First, the central Sahel models have no neighbour histories, so a space-time field at this scale can also absorb diffusion of events between neighbouring cells. In West Africa, adding the space-time field lowers the coefficient of conflict in neighbouring cells from 0.129 (SE 0.007) to 0.051 (SE 0.008) (Table 6). The fall in $\hat{\delta}$ therefore cannot distinguish persistent latent risk from spatio-temporal diffusion. Second, in S3 the empirical SD of $\hat{\delta}$ was 1.19 to 1.66 times its mean SE whenever a space-time field was fitted, so the SEs above are too small, by an

amount that varies across settings. Third, the unrest dynamics are in the regime of S3 (ϕ about 0.9 per week, $\Delta = 8$) in which $\hat{\delta}$ kept an upward bias of about 0.10 under pure heterogeneity. The field v is also defined on the observed cells only, with reflecting edges, which departs from Remark 1; in the space-time hybrid models the same holds for the static part. $\hat{\delta}$ under v should be read with that caveat. As in S3 (Section 4.5), we read the remaining $\hat{\delta}$ as conditional dependence of recorded events (Section 3.4), not as contagion.

Climate.. All models condition on the event histories of the previous four weeks and on the latent effects. The climate coefficients are therefore short-run direct associations. Any effect of climate that acts through earlier events is blocked, and we do not estimate total effects. In the model with the field, the coefficients of the five standardised anomalies for conflict are 0.015 (SE 0.006) for maximum temperature, -0.007 (SE 0.006) for minimum temperature, -0.003 (SE 0.005) for precipitation, -0.004 (SE 0.004) for drought and 0.005 (SE 0.007) for $PM_{2.5}$. For unrest they are 0.009 (SE 0.007), -0.021 (SE 0.006), -0.017 (SE 0.005), -0.006 (SE 0.004) and 0.033 (SE 0.007). The unrest coefficients of minimum temperature and precipitation are thus 3 to 4 SE from zero and larger in absolute value than that of maximum temperature. These SEs treat cell-weeks as conditionally independent given the static latent effects, and residual space-time dependence would make them too small, so we do not read the conflict coefficient of maximum temperature, 2.5 SE from zero, as clear evidence. The $PM_{2.5}$ anomaly is not interpretable. $PM_{2.5}$ is missing in 55.5% of training rows and its climatological standard deviation is undefined in 91,392 of the 164,628 cell-months, so the floor is about 10^{-9} and standardised anomalies can be extreme (Section 2.3). Of the five, we examine the maximum-temperature anomaly in more detail. The biome-specific slopes of the model with a spatially varying coefficient lie between -0.16 and 0.08 for conflict and between -0.09 and 0.06 for unrest, and none is more than two standard errors from zero. The spatially varying part has standard deviation 0.076 for conflict and 0.072 for unrest, so variation within biomes is negligible, and the model does not improve the marginal likelihood or the test scores over the model without it. Conditional on recent event history and the latent effects, the short-run direct association of maximum-temperature anomalies with the weekly probability of recorded events is therefore small in the AME region, and we find no evidence that it differs between or within biomes. This statement does not extend to the other anomalies, nor to total effects of climate.

National indicators.. Without latent effects, several between-country coefficients are large and estimated with standard errors below 0.02, for example -0.56 (SE 0.008) for the EPI on conflict and 0.12 (SE 0.005) for the liberal democracy index on unrest. With country effects their standard errors grow by an order of magnitude, and with the field they change further: -0.22 (SE 0.12) for the EPI on conflict and 0.21 (SE 0.14) for the liberal democracy index on unrest. They are also unstable under spatial+: for the EPI on conflict -0.23, -0.22 and -0.07 with $m = 10$, 20 and $m^* = 138$, and for the liberal democracy index on unrest 0.17, 0.20 and -0.02 with $m = 10$, 20 and $m^* = 176$. In the light of S2 (Section 4.4), the between-country coefficients cannot be separated from spatially smooth unmeasured risk, and we do not interpret them.

The within-country coefficients are identified from changes over time within a country. Across the models with the field (with the field, with the spatially varying coefficient, and spatial+), they barely change, for example 0.014 to 0.016 for the EPI on conflict and 0.030 for the liberal democracy index on unrest. They are not all small. In the model with the field, the within coefficients on conflict are 0.095 (SE 0.007) for log GDP per capita, -0.082 (SE 0.011) for greenhouse gas emissions and 0.049 (SE 0.010) for the ecological footprint, several times the maximum-temperature coefficient. Their standard errors, however, treat cell-weeks as conditionally independent, whereas the within variation comes from about 93 countries

over five training years. Like the between coefficients, they rest on few independent contrasts, and their model-based standard errors are too small by an unknown amount. We therefore do not interpret them either. Access to electricity needs a further caveat. Unlike the other national indicators, it is not re-attached per country and year in the repaired table (Section 2.1), so in cells that had copies it can carry the value of a neighbouring country. Its within component then mixes change over time with differences across borders. This may explain why its within coefficient on conflict is -0.106 (SE 0.005) without latent effects but -0.027 (SE 0.023) with the field, and on unrest 0.008 (SE 0.006) without latent effects but 0.072 (SE 0.024) with the field. We do not interpret the electricity coefficients.

Missingness.. Two missingness indicators were dropped because they are constant in the training rows (EPI and liberal democracy index), and the material and ecological footprint indicators were merged with that of greenhouse gas emissions. The remaining indicators are not interpretable as effects. They vary mostly between countries, so they are confounded with the country effects and the field, and their signs change across specifications. For a missing GDP the coefficient on conflict is 0.31 (SE 0.022) without latent effects, -0.08 (SE 0.061) with country effects and -0.26 (SE 0.056) with the field. For a missing IPC analysis the coefficient on unrest is 0.06 (SE 0.020), 0.20 (SE 0.023) and -0.18 (SE 0.030) in the same three models. The other indicators have large standard errors. [TODO: National-indicator coefficients refitted on the countries with complete indicators.]

6. Discussion.

[TODO: Discussion (not drafted yet).]

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SUPPLEMENTARY MATERIAL

Code

Python implementation of the latent Gaussian model, the simulator and all simulation studies.

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